

Support Package

The data except this specification conform to those of each product.

1. Product type

IMX530 series :

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series :

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series :

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series :

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series :

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series :

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series :

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

This support package is intended to support product development.

3. Description Items

- ◆ Guideline for designing the printed circuit board
- ◆ SLVS / SLVS – EC output interface (Excluding IMX548 series, IMX565 series and IMX568 series)
- ◆ SLVS output interface (IMX548 series)
- ◆ CSI-2 output interface (IMX565 series and IMX568 series)
- ◆ Communication port (4-wire serial / I²C)
- ◆ Pattern Generator (PG) function
- ◆ How to get sensor information
- ◆ Lens design guideline
- ◆ FD white spots

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

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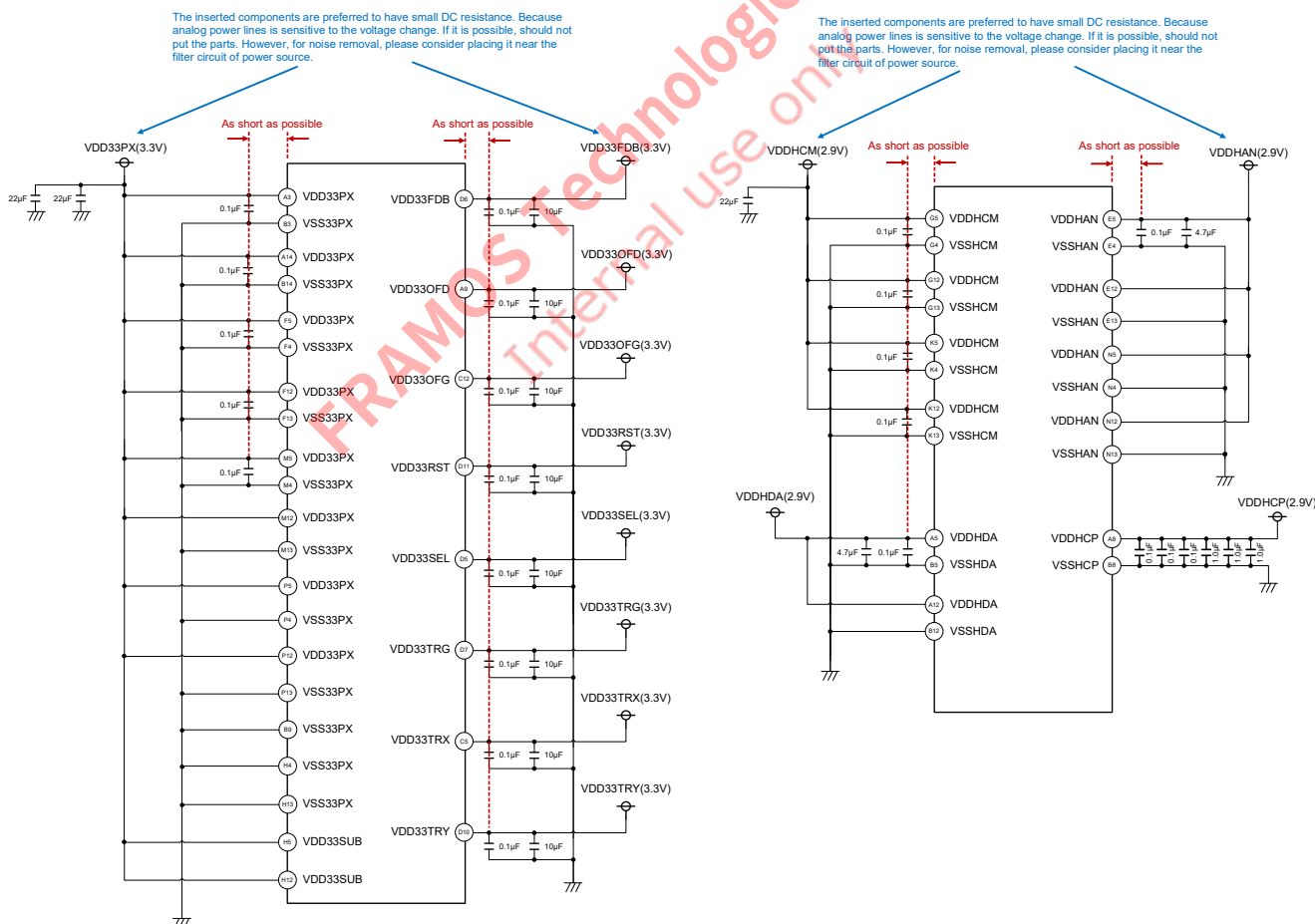
4. Guideline for designing the printed circuit board

In this section we explain the design guideline of the printed circuit board layout and mount.

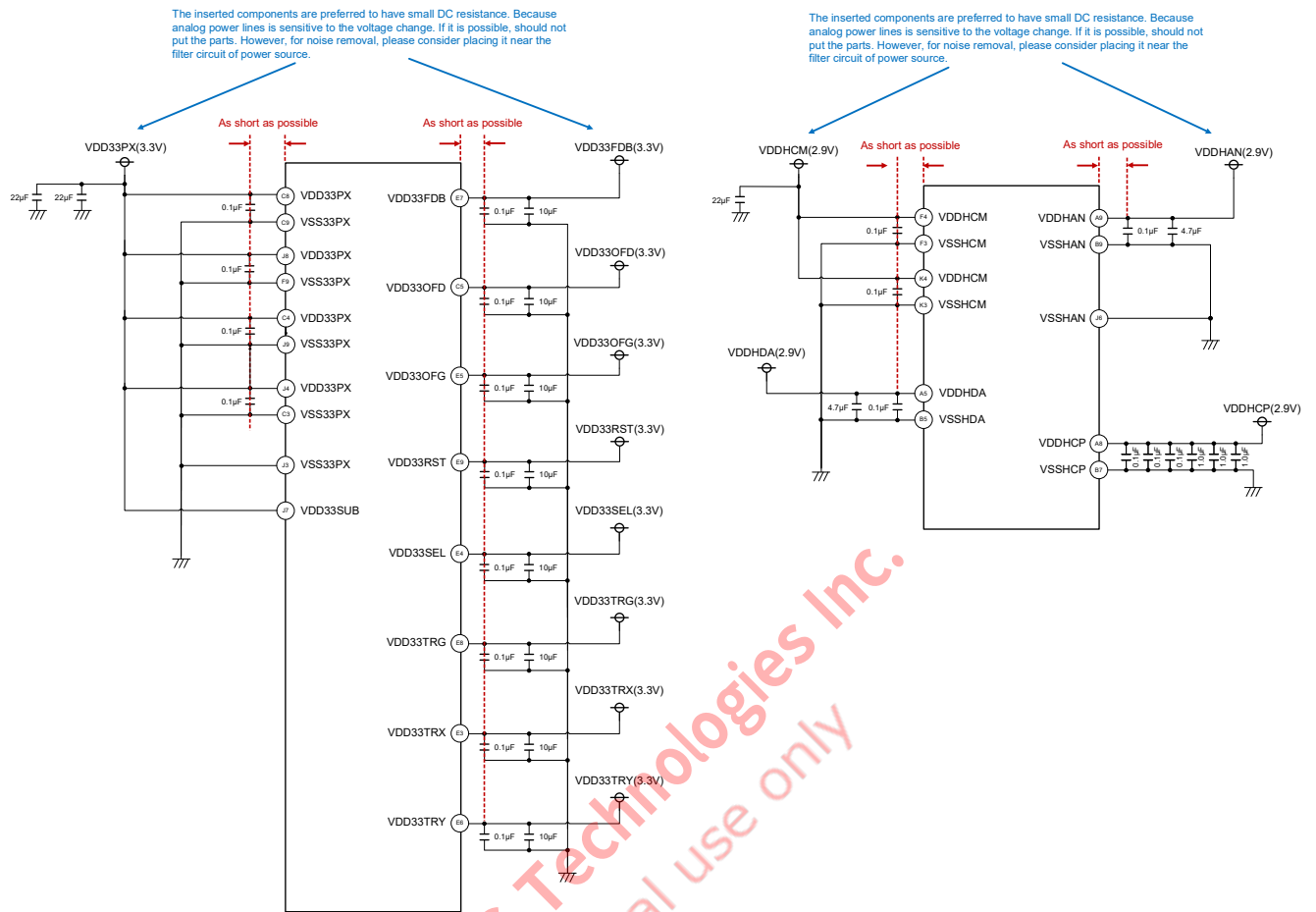
4.1. Application circuit. (LGA package)

4.1.1. Power supply pins

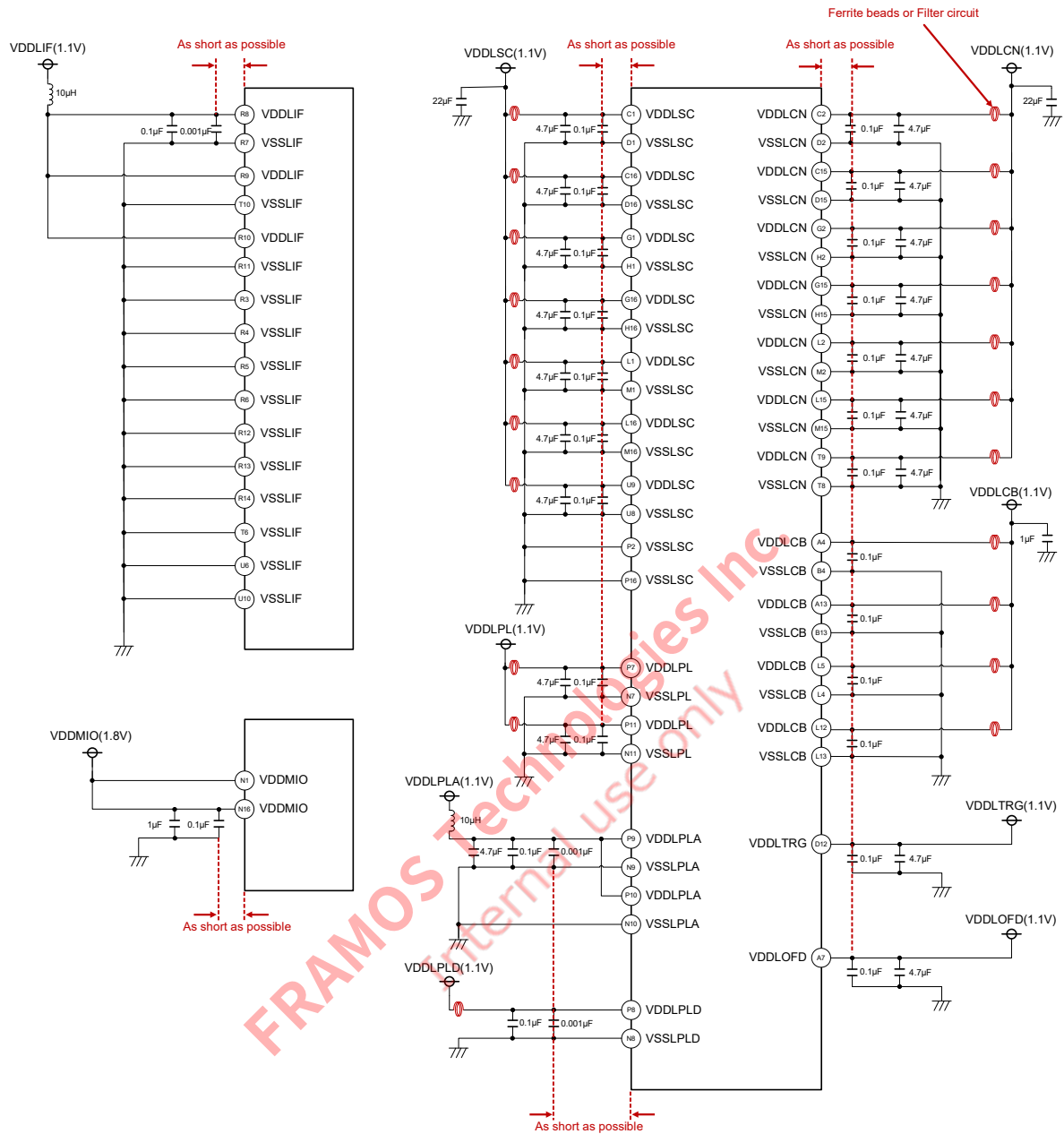
- (1) Please design the suitable filter in power supply line to reduce influence of a power supply noise and to prevent an unnecessary radiation. Especially, the parts to analog power supply pin need to choose the one that is small direct current resistance because that pin is sensitive to the voltage change. The voltage change becomes the factor of horizontal line noise.
- (2) We suggest that the use of reasonable noise filters on the power-line for suppressing the radiation noise from the lines.
- (3) The power supply pins which are sensitive to a noise from the outside are as follows. We recommend it not to become the common impedance with an analog power supply and an other analog power supply (e.g. Separate the wiring from power supply regulators with other analog power supplies).
excluding IMX548 series and IMX568 series :
A3, A14, F5, F12, M5, M12, P5, P12, E5, E12, N5, N12 Pin (Analog 2.9 / 3.3V):
IMX548 series and IMX568 series :
C4, C8, J4, J8, A9 (Analog 2.9 / 3.3 V)
- (4) Please mount peripheral parts (capacitor) near the element as much as possible.
- (5) Please make patterns of Power supply (3.3V, 2.9V, 1.8V, 1.1V) as wide as possible.
- (6) Please put capacitors at each power pin. And avoid putting a capacitor of total capacity value at multiple pins in order to reduce parts.



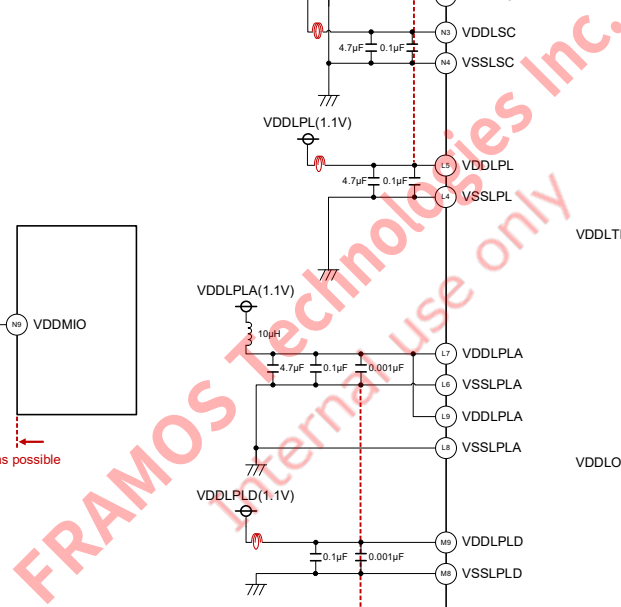
Reference design for analog power supply pins (excluding IMX548 series and IMX568 series).



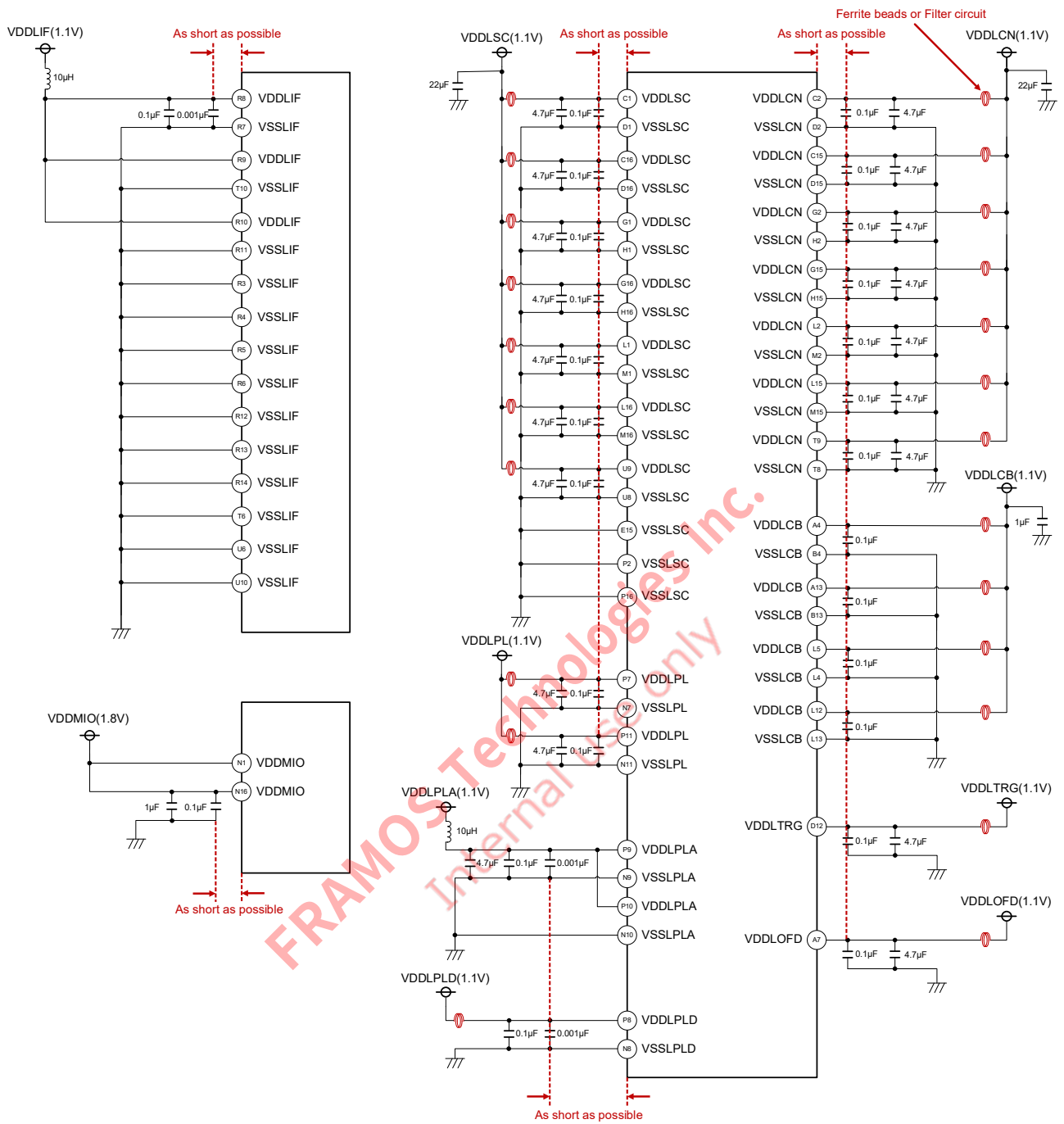
Reference design for analog power supply pins (IMX548 series and IMX568 series).



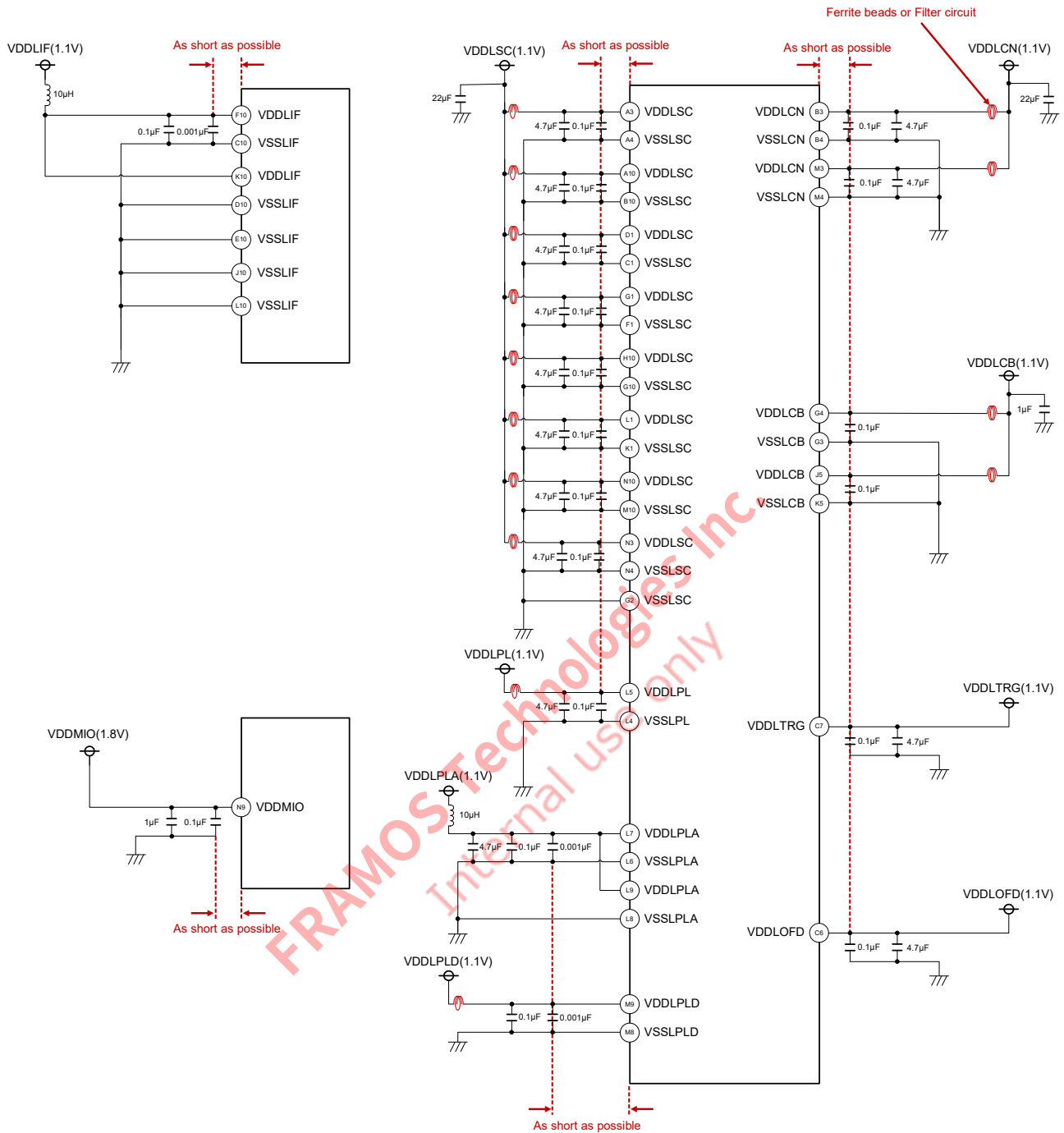
Reference design for digital power supply pins (excluding IMX565 series, IMX548 series and IMX568 series)



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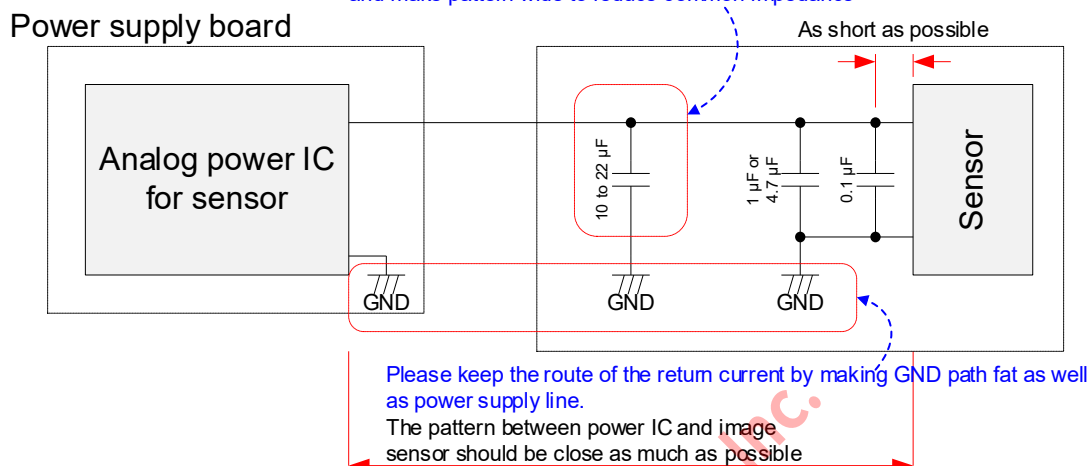


Reference design for digital power supply pins (IMX565 series)



Reference design for digital power supply pins (IMX568 series)

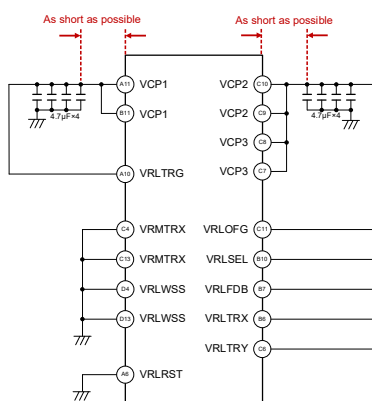
- Bypass capacitor : 10 to 22 μF (If necessary)
It is preferable to put capacitor separately in each power pin
If you bring power line together, please bring it together near the IC,
and make pattern wide to reduce common impedance



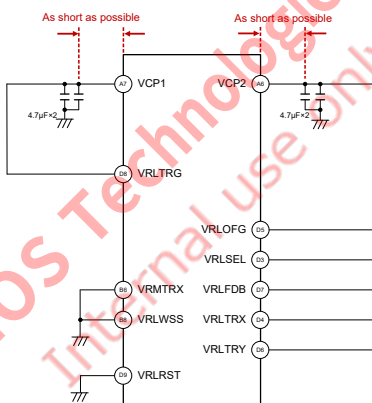
Layout of power supply IC and decoupling capacitors of large capacity

4.1.2. Other Pins

VCP1, VCP2, VCP3 pin is sensitive to noise. Please place the capacitors near the pins. As for the capacity of a capacitor, it is desirable to insert the value according to the example of an application circuit.



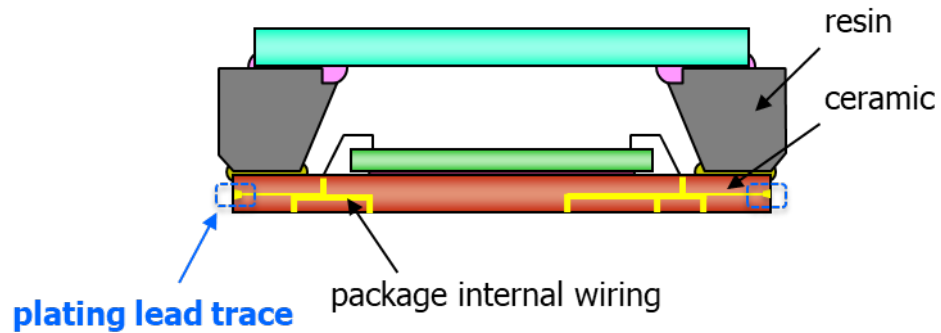
Reference design for other pins. (excluding IMX548 series and IMX568 series)



Reference design for other pins. (IMX548 series and IMX568 series)

4.1.3. Notes for the mount

In the ceramic part of the package, there is a part where the internal wiring appears on the surface as a plating lead trace. When conductors touch the package surface, they will be connected to the package internal wiring. So please do not touch the package surface with conductive material.



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4.2. (Reference) Component

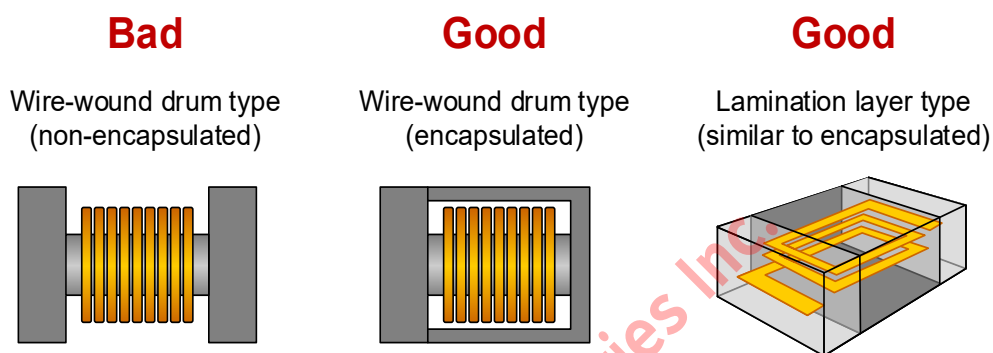
The following describes the components used reference information. When in actual use, please contact the manufacture for each component.

4.2.1. Power supply IC

Please select the power supply IC of better PSRR characteristics. Fluctuation of the power supply voltage will cause the horizontal line noise. In that case, we recommend that you use a series regulator.

Please note that when using switching power regulator of the following points.

- (1) When using switching power regulator, magnetic field induced by inductor might cause the noise of the image.
- (2) Encapsulated (shield) type inductor is recommended.
- (3) Mounting direction of the inductor also affects (increase or decrease) the noise level.

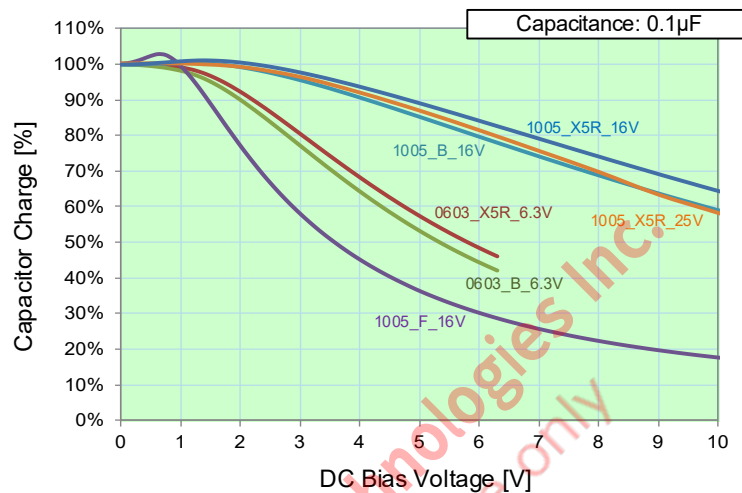


Encapsulated/non-encapsulated type inductor

4.2.2. Decoupling capacitors

Please use non-polar capacitors. It is possible to operate the sensor by using typical ceramic capacitors.

- (1) All the power supply voltage are smaller than 3.3 V so you can use any 1608M size capacitors of any characteristics in the graph. When you use 1005M size capacitors, you should select characteristics-B device. Characteristics-F device should not be chosen.
- (2) Decoupling capacitors for each pin are sensitive to image quality. Please implement sufficient capacity of the decoupling capacitors according to the power supply condition. If you use capacity that is small one and large one, please put the small one near the sensor. Please refer to Application circuit for details.
- (3) The capacity of the laminated layer ceramic capacitors will decrease in regard to the impressed DC bias voltage by piezo-electric effect. It will be more obvious when using smaller size and larger capacity, please check the specification sheet of each devices.
(Graph below shows the characteristics.)

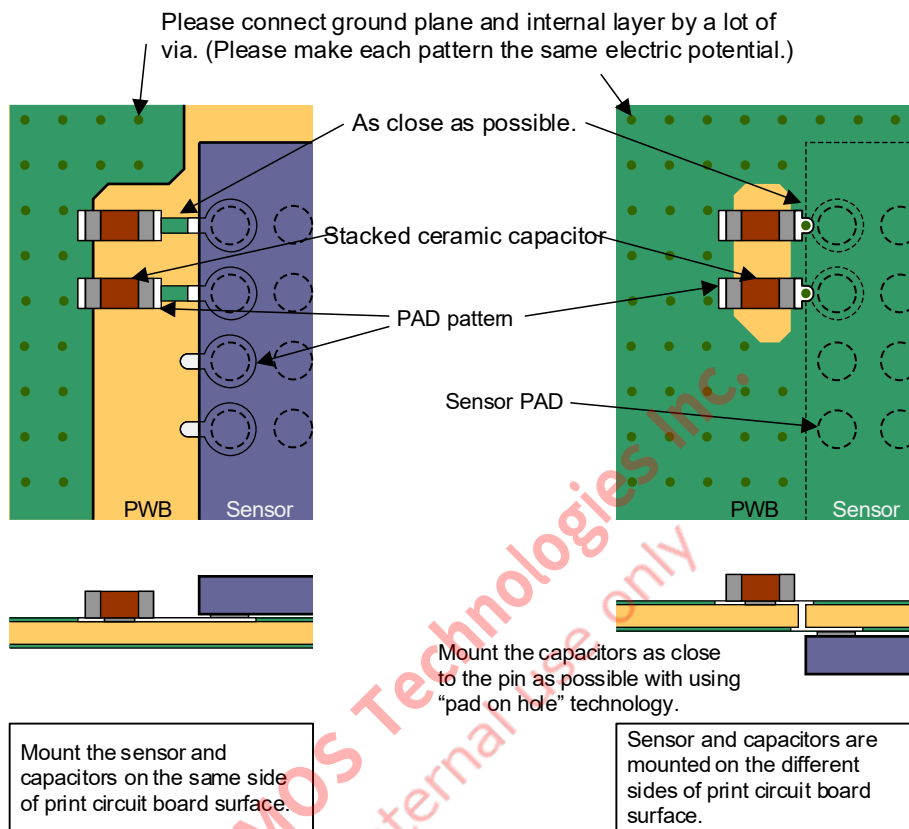


DC bias vs. capacity

4.3. Notes for designing patterns of printed circuit board.

4.3.1. Power supply pins

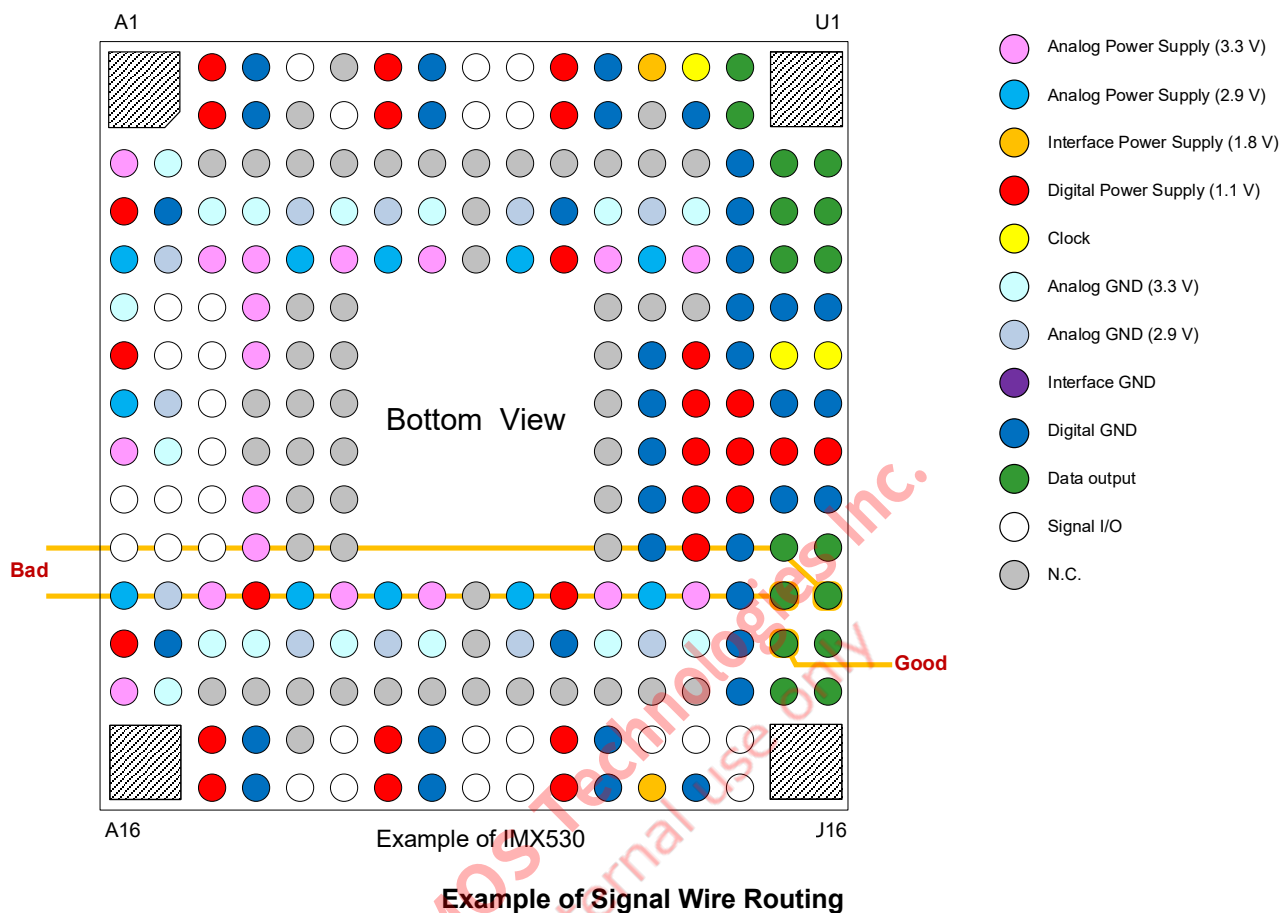
- (1) The peripheral devices (capacitors) of the sensor should be mounted as close to the power supply pins as possible. Longer wire length might cause the degradation of the image quality.
- (2) Surface power/ground plane and inner layer power/ground plane should be connected with enough number of the vias (Vertical Interconnect Access) . Please make each pattern the same electric potential.



Example of capacitor mounting

- (3) The power supply pins which are sensitive to a noise from the outside are as follows. We recommend it not to become the common impedance with an analog power supply and an other analog power supply (e.g. Separate the wiring from power supply regulators with other analog power supplies).
excluding IMX548 series and IMX568 series :
A3, A14, F5, F12, M5, M12, P5, P12, E5, E12, N5, N12 Pin (Analog 2.9 / 3.3V):
IMX548 series and IMX568 series :
C4, C8, J4, J8, A9 (Analog 2.9 / 3.3 V)
- (4) Loop of the ground pattern might propagate the noise to the other patterns (ex. Power supply pattern) so the ground pattern recommends Land pattern. Its recommended to use the common single GND plane for Analog GND and Digital GND.

- (5) Digital signal pattern on the other layer should not be in parallel with the power supply patterns, terminals of VCP1, VCP2, VCP3, VRLWSS, VRLSEL, VRMTRX, VRLTRX, VRLTRY, VRLOFG, VRLTRG, VRLRST, VRLFDB and these lines.



4.4. Wiring patterns for image output signals

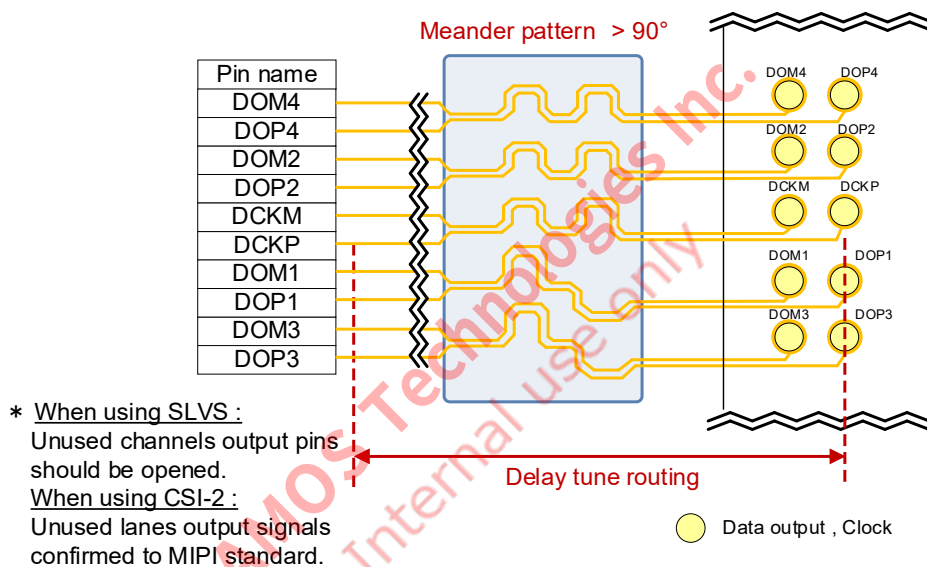
This sensor is supported output format below. This section described about wiring pattern for each output format.

- ◆ SLVS output (Excluding IMX565 series and IMX568 series)
- ◆ SLVS – EC output (Excluding IMX548 series, IMX565 series and IMX568 series)
- ◆ CSI-2 serial output (IMX565 series and IMX568 series)

4.4.1. SLVS (Excluding IMX565 and IMX568 series) / CSI-2 (IMX565 and IMX568 series) serial output

- (1) Signal wires (DMOxP and DMOxN, DCKP and DCKN) must be paired. (Differential impedance is 100 Ω typ.)
- (2) We recommend delay tune wiring for image signals, especially delay of each differential pair signal and Data and its Strobe signal should be controlled by using meander wiring. Turning point angle of the wire should be greater or equal to 90 degree. (obtuse angle)
- (3) Decoupling capacitors for power supply for image signal output (VDDLIF) should be mounted close to the power supply pins of the package of the sensor with using small size (1005M or 0603M) laminated layer ceramic capacitor.

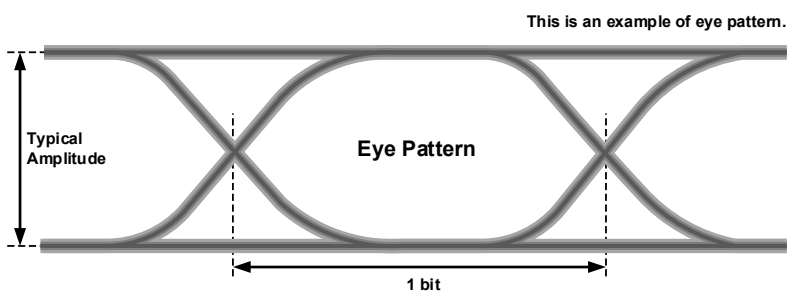
Meander pattern should make pattern as close to the output pins (near the sensor) as possible.



Example of wiring patterns for SLVS output signal

<For your reference>

Validity of the image signal can be checked by “eye pattern” of the signals. With the adequate loading and delay tuned condition, you can see clear “eye pattern” with sufficient margin of the threshold and the noise level for the receiver.



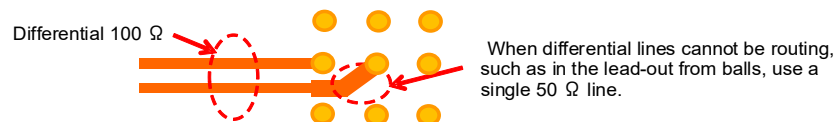
Eye pattern of the image signal

4.4.2. SLVS – EC output (Excluding IMX548 series, IMX565 series and IMX568 series)

(1) transmission line

Use the line and space values guaranteed by the board manufacturer for a differential of 100 Ω . The allowable impedance tolerance is $100 \pm 10 \Omega$.

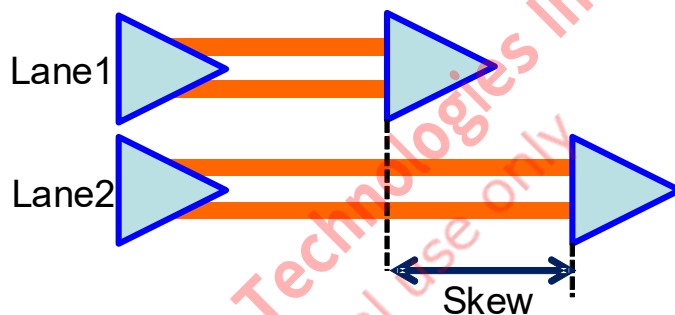
In locations where differential lines cannot be routed, route using the line and space values for single 50 Ω (BGA ball edges, etc.) Use the routing layout that results in the shortest routing possible.



When bending a transfer path, do not bend the routing at 90°. Instead, use 45° routing or establish a bending radius (R).



Differential pair (positive/negative) lines should use equal-length routing (routing length difference: 1 mm or less). There is no need to make each lane equal length, but see the SLVS – EC Specification for the allowable skew.



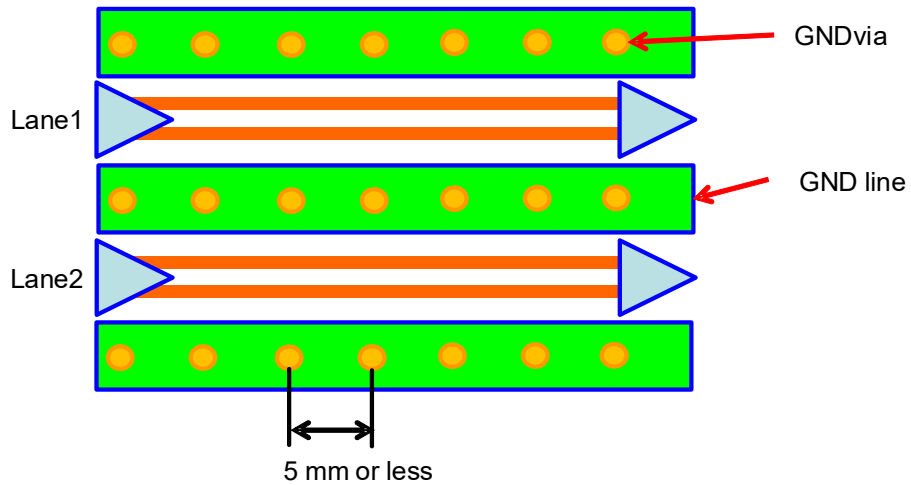
(2) GND Line

Insert a GND line between differential lanes to reduce crosstalk.

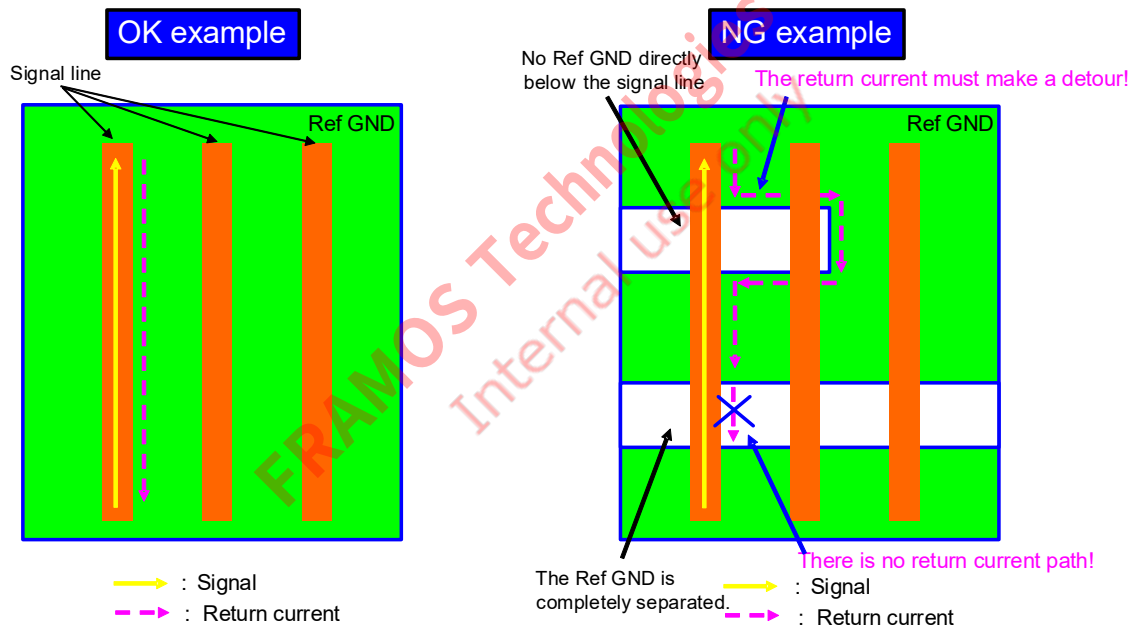
When it is physically impossible to insert a GND line, separate the lanes. (Guideline: 3 times the lane width)

(3) Through-hole Placement

Locate through-hole via (GND via) that connect to the solid GND on the L2 layer at intervals of 5 mm or less in the GND lines inserted in 2 above.



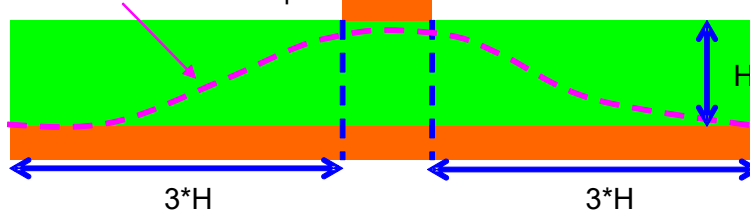
(4) GND Plane to be Secured for the Return Current Path



(5) Ref GND Plane

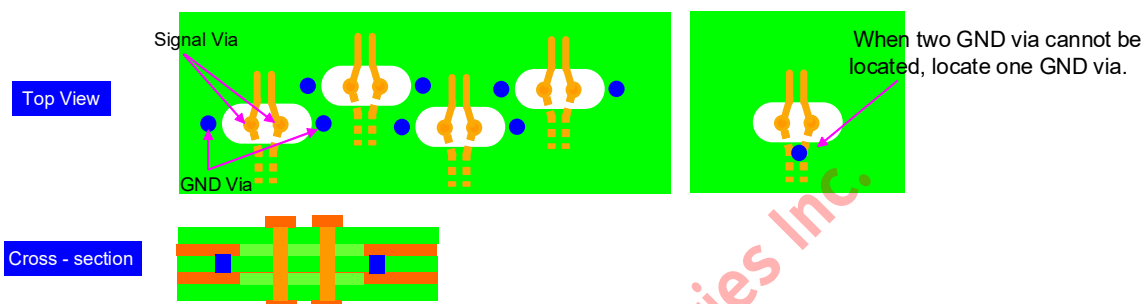
Secure a GND plane on both sides for 3 times or more the thickness (H) of the insulator (guideline).

This becomes the return current path.



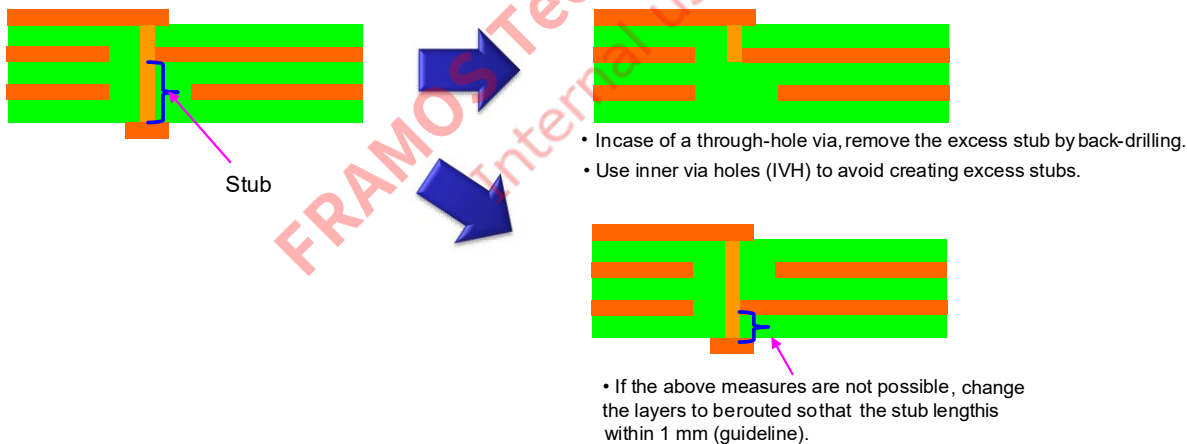
(6) GND Via

When it is necessary to route a signal line to a different layer through a via, locate GND via for the return current.



(7) Signal Via and GND Via

When it is necessary to route a signal line to a different layer through a via, if at all possible arrange the signal via and GND via in a manner to avoid creating a stub. In the event that a stub is created, make sure that the stub length is within 1 mm (guideline).



(8) INCK Routing

Use the routing layout that results in the shortest routing possible.

The Ref GND should be a solid GND.

Arrange the routing so that there is no crosstalk between analog power supplies, signals and INCK.

Separate routing by 3 times or more the width of the signal lines or shield with GND to prevent signal lines from running parallel with other routing.

(9) Notice for development at SLVS – EC (Baud Grade 3)

When designing SLVS – EC, care should be inconsistent at impedance.

In designing systems, use Tx IBIS – AMI when simulation for confirm to fast – signals.

For development SLVS – EC, we can send following documents.

(*If sending these documents, you will must be sign to License Agreement.)

- SLVS – EC Specification*
- SLVS – EC Application note*
- Tx/Rx C model*
- Conformance Test Suite*
- SLVS – EC Board Design Guidelines for SLVS – EC Interface
- Tx IBIS – AMI model (for Baud Grade 3) or IBIS model (for Baud Grade 1, 2) for PHY simulation

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5. Output Signal Format

This section describes the operation of output mode supported by this sensor.

5.1. SLVS – EC Output (Excluding IMX548 series, IMX565 series and IMX568 series)

SLVS – EC embeds the clock to the stream of data.

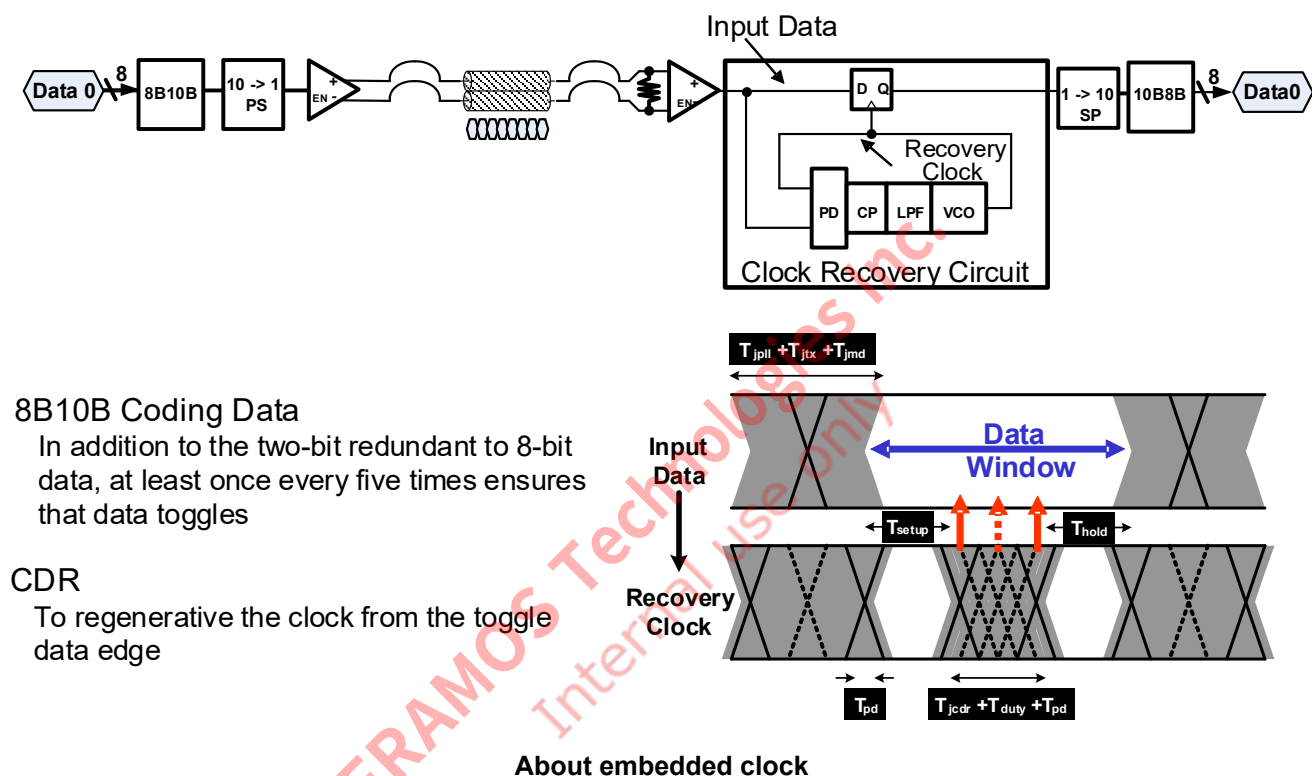
As a result, it was made unnecessary the skew adjustment troubled by high-speed interface.

The outline / For further information of SLVS – EC standard, please refer the following document.

- SLVS – EC Specification

5.1.1. Embedded Clock

For embedding the clock in the data, skew adjustment is not required.



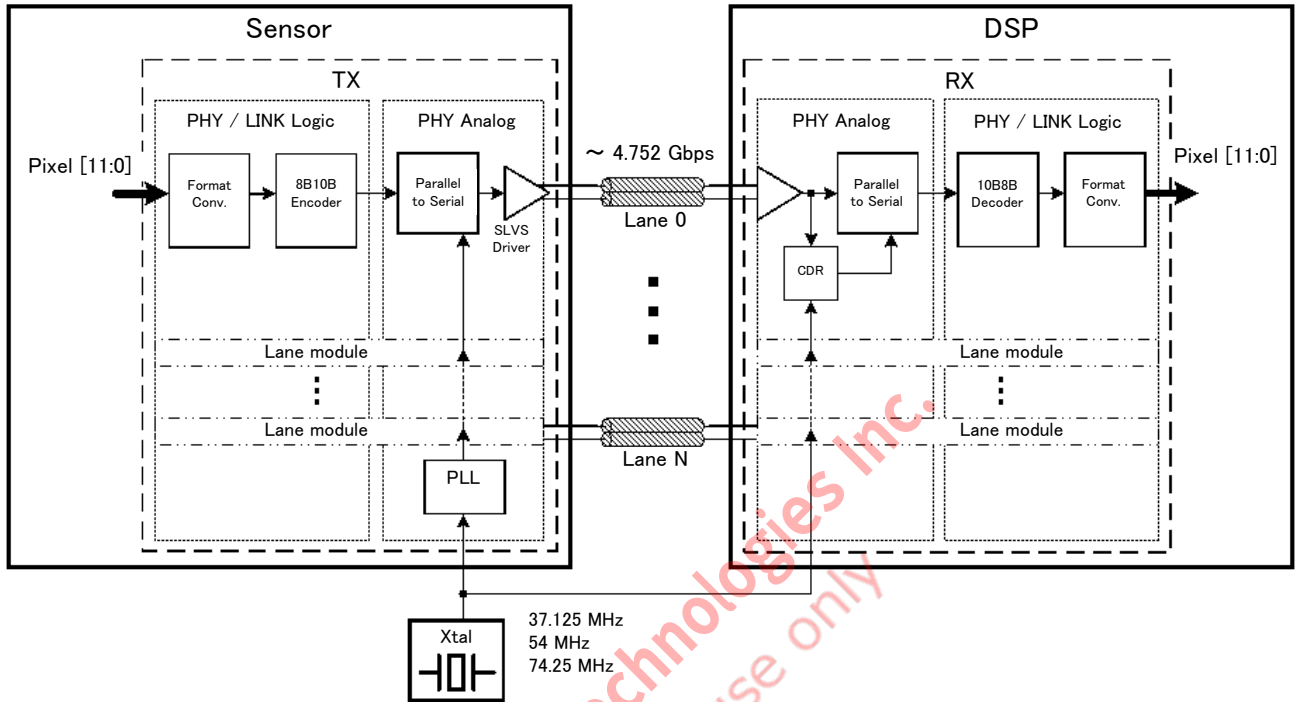
5.1.2. SLVS – EC Block chart

SLVS – EC block is composed of LINK Logic, PHY Logic, PHY Analog.

LINK Logic: To convert the pixel information in Byte units.

PHY Logic: To make the protocol control · 8B10B conversion and synchronization code addition.

PHY Analog: To serialized and output in SLVS level.



5.2. CSI-2 Serial Output (IMX565 series and IMX568 series)

For details of MIPI Standard, output format, packet header and footer, please refer followings.

- MIPI Alliance Specification for Camera Serial Interface 2 (CSI-2) Version 1.2
- MIPI Alliance Specification for D-PHY Version 1.2

6. Serial Communication Port

This sensor has 4-wire serial communication and I²C serial communication.

This section describes the operation of the serial interfaces.

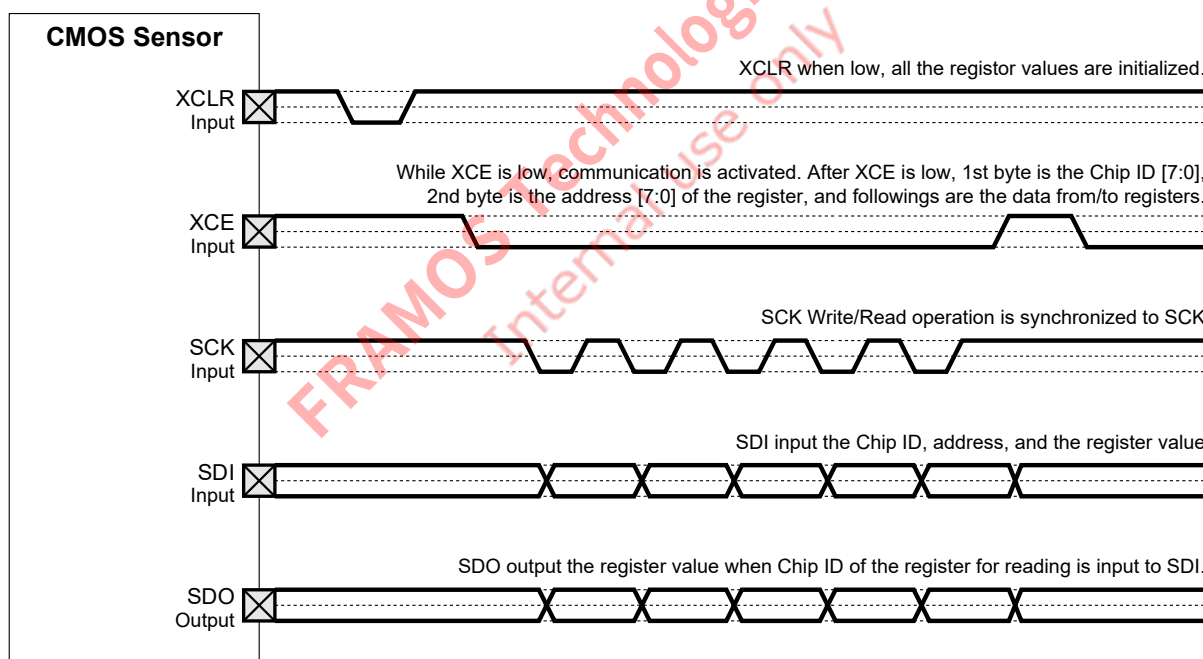
6.1. 4-wire serial communication

Characteristics of 4-wire serial interface.

- (1) 3-wire serial interface consists of serial data input (SDI), serial clock (SCK), and chip enable (XCE). SDO is the data output port for read out the value in the registers.
- (2) Data transfer is done in unit of 8bit (1byte). In case of continuous communication, no limitation for the numbers of byte for the communications. (However, limited by communication period.)
- (3) LSB first protocol.

Ex.) When sending 28h LSB first: 0001 0100.

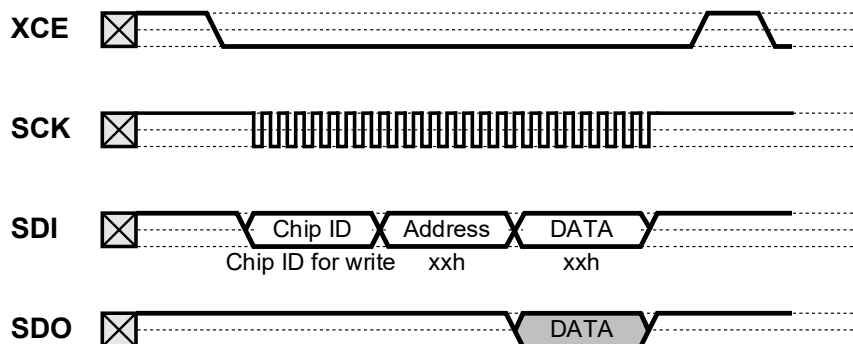
Terminal name	function	description
XCLR	Initialize the system	Initialize the register value.
XCE	Enable the communication	Communication is active for XCE is low.
SCK	Serial clock for communication	Maximum frequency is 13.5 MHz
SDI	Serial data input	Change the value at falling edge of SCK, latch the value by rising edge
SDO	Serial data output	Output the value synchronous to the falling edge of SCK



4-wire Communication Function

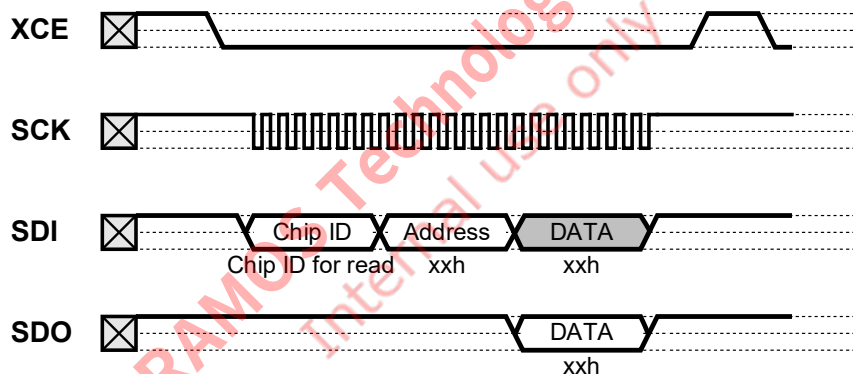
6.1.1. Register write/read operation

Both of register write and read operation, Chip ID, address and register value are declared to SDI. When Chip ID is the registers for writing, value is written to the register designated by the address. When Chip ID is the registers for reading, the value is read out from the register designated by the address. Write and read operation can be done continuously while XCE is low.



Write Operation

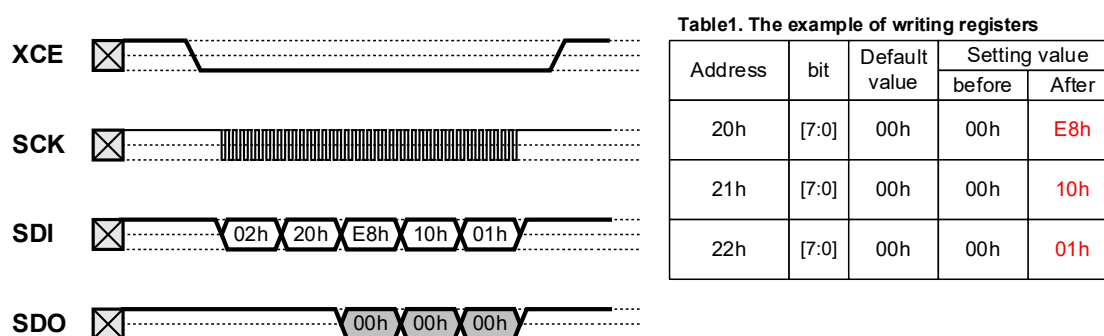
When write operation, the order of the commands to SDI is as follows. 1st byte: Chip ID is selected the registers for writing, 2nd byte: address to write, 3rd byte and after: register value. SDO output the current value (Before write) of the register designated by the address.



Read Operation

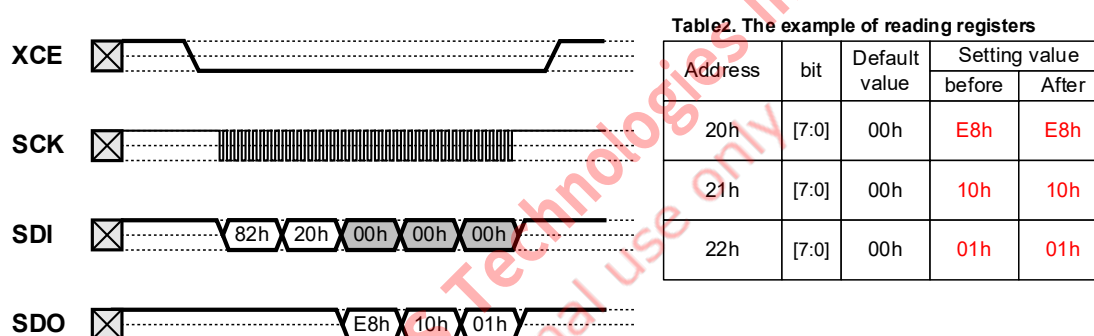
When read operation, the order of the commands to SDI is as follow. 1st byte: Chip ID is selected the registers for reading, 2nd byte: address to read, 3rd byte and after : invalid data SDO output the value of the register designated by the address.

The example of communication timing to writing to continuous address as shown table 1 is shown below.



Write operation to continuous address

Write E8h to address 20h first. And with keeping XCE low, by writing 10h, 01h in succession, these values are set to consecutive addresses (21h and 22h). The example of reading communication timing to writing to continuous address as shown table 1 as shown table 2 is shown below.



Read operation to continuous address

In case of reading out the value from the registers, write the registers for reading as the Chip ID then designate the address to read. After that input the dummy data to SDI. In read mode, input data are discarded and no register is updated, current register values are output from SDO continuously.

6.2. I²C serial communication

Characteristics of 4-wire serial interface.

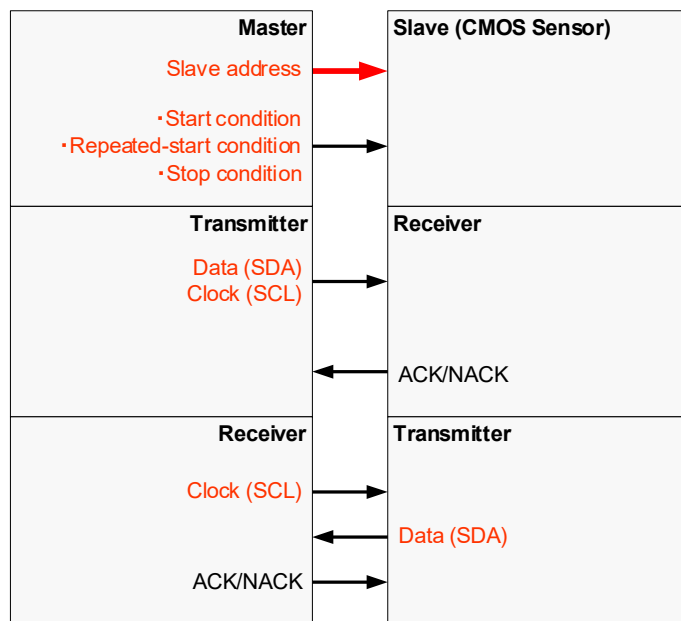
- (1) I²C consists of 2 wires of serial data and serial clock.
- (2) Unit of the communication is 8 bits (1 byte), bi-directional communication. And in case of continuous communication, no limitation for the numbers of byte for the communications. (However, limited by communication period.) Acknowledge bit (ACK/NACK) is required at the end of communication.
- (3) MSB first protocol
e.g.) When sending 28h MSB first: 0010 1000.
- (4) I²C is the bi-directional communication and the "master" side controls the communication, the "slave" side is controlled by master. The "master" always provides the clock.
- (5) While communication, the device transferring the data is the "transmitter", and the device receiving the data is the "receiver". The "receiver" provides acknowledge bit.
- (6) This sensor is slave.

I²C Communication mode for this sensor

Mode	Data rate	Correspondant
Standard mode	100 kbps	Available
Fast mode	400 kbps	Available
Fast mode Plus	1 Mbps	N/A
High speed mode	3.4 Mbps	N/A

SLAVE Address

Refer to the data sheet of the image sensor.



Relation between the sensor and the ISP when I²C communication.

Master execute the following communication.

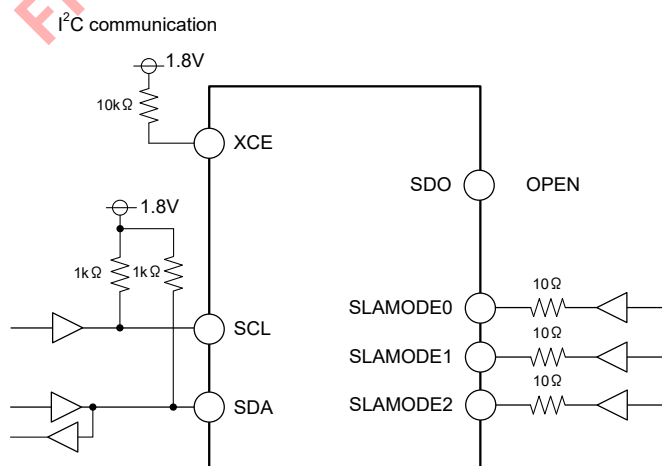
- ◆ Generate slave address (declare the device to be slave. Sensor is the slave here.)
- ◆ Generate "start condition", "repeated start condition", and "stop condition".

While communication, the device transforming the data is the "transmitter", and the device receiving the data is the "receiver". Master can work as both transmitter and receiver, slave also work as both.

Transmitter generates the data (SDA), and receiver generates the acknowledge Bit (ACK/NACK).

Master always generates the clock (SCL).

Reference design for I²C serial communication is shown below. Please connect XCE and OV_{DD} when I²C communication. In addition, SCL, SDA, please pull up to 1 k Ω OV_{DD}.



Reference design for I²C serial communication

6.2.1. Overview of the communication protocol

Data is transferred by SDA port. The transition timing of the SDA follows the rule below.

- ◆ State control (start, restart, stop of the communication) is done while SCL is "high".
- ◆ When data transfer, SDA toggles while SCL is "low"

6.2.1.1. State control (start, restart, stop of the communication)

The conditions of start (start condition), restart (repeated start condition), and stop (stop condition) of the communication is shown below. State control is done by master device.

State condition	condition
Start condition	SDA toggles from High to Low while SCL is high
Repeated start condition	start condition while stop condition is of the previous data transfer is not Declared
Stop condition	SDA toggles from Low to High while SCL is High.

6.2.1.2. Acknowledge bit

Data transfer is done in unit of 8 bits (1 byte). The acknowledge bit is generated for every 8 bits data transfer and added after the last (8th) bit to indicate that receiver normally received the data. When receiver wants to stop the communication by the internal interrupt or etc. receiver generates the negative acknowledge bit (NACK).

- ◆ Master must send the slave address after declaration of the "Start condition" and "Repeated start condition".
- ◆ When slave generates the negative acknowledge bit, master declares the stop condition and stops the communication immediately.

7. Note for access to register

When writing the value to registers, we recommend "Read-modify-write" to avoid the trouble caused by inadequate over write of the register. For example "VREVERSE" value is assigned to address of 04h [0], "HREVERSE" value is assigned to address of 04h [1]. When update HREVERSE value after writing VREVERSE, VREVERSE value in address of 04h [0] should be kept. "Read-modify-write" can prevent the wrong overwrite of the value

Ex.) In case of writhing the value 1d to VREVERSE and HREVERSE

In case of "no Read-modify-write"

- 1) HREVERSE = 1d ... set value 02h to address: 04h

Address: 04h

0	0	0	0	0	0	1	0
---	---	---	---	---	---	---	---

- 2) VREVERSE = 1d ... set value 01h to address: 04h

01h

Address: 04h

0	0	0	0	0	0	0	1
---	---	---	---	---	---	---	---

HREVERSE=0d VREVERSE=1d

Writing the VREVERSE value 01h to address 04h without considering HREVERSE bit in 04h [1]. HREVERSE will be changed.

In case of "Read-modify-write"

- 1) HREVERSE = 1d ... set value 02h to address: 04h

Address: 04h

0	0	0	0	0	0	1	0
---	---	---	---	---	---	---	---

- 2) Read register value

0	0	0	0	0	0	1	0
---	---	---	---	---	---	---	---

Read out the current value and store the value in the memory area of MCU

- 3) VREVERSE = 1d ... set value 01h to address: 04h

01h

Address: 04h

0	0	0	0	0	0	1	1
---	---	---	---	---	---	---	---

Bit operation "add" with the value stored in step 2)

03h

Address: 04h

0	0	0	0	0	0	1	1
---	---	---	---	---	---	---	---

HREVERSE=1d VREVERSE=1d

Write value after bit operation (No overwrite of VREVERSE value)

8. Pattern Generator

This chapter explains the pattern generator (PG) function.

8.1. Register map for pattern generator function

The Register map for Pattern Generator is shown below.

Please set via sensor standby.

Please refer to the datasheet for registers setup other than those lists.

Chip ID = 07 (Write: Chip ID = 07h, Read: Chip ID = 87h, I²C: 35h)**

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
20h	3520h	0		Fixed to 1Fh	10h	10h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
40h	3540h	0		Fixed to 00h	01h	01h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
50h	3550h	0	PGREGEN [0]	PG operation enabled 0h : PG OFF 1h : PG ON	0		06h ^{*1} 02h ^{*2}	S
		1	PGTHRU [0]	Back Ground Transient 0h : Invalid PGMODE = 00h / 06h; The background is fixed to "0h". PGMODE = 04h / 05h; The background is set to PGDATA2. 1h : Valid	1			S
		2	PGCLKEN [0]	Clock control for PG 0h : Clock stop 1h : Clock operation Set to "1h" when using Pattern Generator	1 ^{*1}	0 ^{*2}		S
		3		Fixed to 0	0			—
		4		Fixed to 0	0			—
		5		Fixed to 0	0			—
		6		Fixed to 0	0			—
		7		Fixed to 0	0			—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
51h	3551h	0	PGMODE [4:0]	PG mode setting	00h	00h	S
		1		00h: Multiple pixels Pattern			
		2		01h: Sequence Pattern 1			
		3		02h: Sequence Pattern 2			
				03h: Gradation Pattern			
				04h: Horizontal 1 Row Pattern			
				05h: Vertical 1 Column Pattern			
				06h: Horizontal 1 Row and Vertical 1 Column Pattern			
		4		07h: Stripe Pattern of the arbitrary value			
				08h: Checks pattern of the arbitrary value			
				0Ah: Color bar which changes horizontally			
				0Bh: Color bar which changes vertically			
				Others: Setting prohibited			
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
54h	3554h	0	PGHPOS [12:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6		Horizontal Start Address			
		7		0 or more is valid.			
55h	3555h	0	PGVPOS [12:0]	MSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
56h	3556h	0	PGVPOS [12:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6		Vertical Start Address			
		7		0 or more is valid.			
57h	3557h	0	PGVPOS [12:0]	MSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
58h	3558h	0	PGHPSTEP [7:0]	Interval of horizontal pixels 00h is prohibited	01h	01h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
59h	3559h	0	PGVPSTEP [7:0]	Interval of vertical pixels 00h is prohibited	01h	01h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
5Ah	355Ah	0	PGHPNUM [7:0]	Number of horizontal pixels 00h is prohibited	01h	01h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
5Bh	355Bh	0	PGVPNUM [7:0]	Number of vertical pixels 00h is prohibited	01h	01h	S	
		1						
		2						
		3						
		4						
		5						
		6						
		7						
5Ch	355Ch	0	PGDATA1 [12:0]	LSB	0000h	00h	S	
		1						
		2						
		3						
		4						
		5						
		6		Set PGDATA1				
		7						
5Dh	355Dh	0		MSB	0	00h		
		1						
		2						
		3						
		4						
		5		Fixed to 0				
		6		Fixed to 0				
		7		Fixed to 0				

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
5Eh	355Eh	0	PGDATA2 [12:0]	LSB	0000h	00h	S
		1					
		2					
		3					
		4					
		5					
		6		Set PGDATA2			
		7					
5Fh	355Fh	0			0	00h	—
		1					
		2					
		3					
		4		MSB			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
60h	3560h	0	COLORWIDTH [7:0]	LSB	0Bh	0Bh	S
		1					
		2					
		3		Color bar width setting			
		4		16 pixels unit			
		5		00h and 01h are prohibited			
		6					
		7		MSB			
62h	3562h	0	PGHGSTEP [1:0]	The increment of gradation pattern	0h	00h	S
		1		00b: +1 / 01b: +2 / 10b: +4 / 11b: prohibited			
		2		Fixed to 0			
		3		Fixed to 0			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			
B4h	35B4h	0	BLKLEVEL [11:0]	LSB	03Ch	3Ch	V
		1					
		2					
		3					
		4		Black level offset value setting			
		5		Set to "000h" when using Pattern Generator			
		6					
		7					
B5h	35B5h	0			0	00h	—
		1					
		2					
		3		MSB			
		4		Fixed to 0			
		5		Fixed to 0			
		6		Fixed to 0			
		7		Fixed to 0			

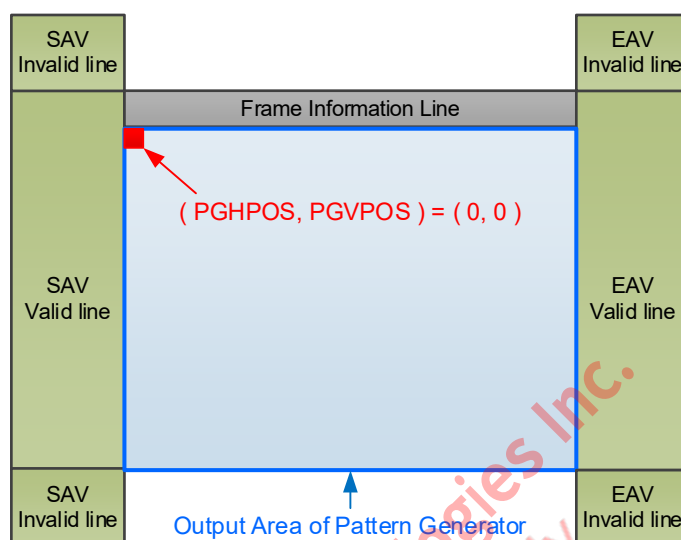
¹ IMX530 series and IMX540 series.

² IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

8.2. Common setting item for the PG

8.2.1. When SLVS (Excluding IMX565 series and IMX568 series) output

The output area of the PG pattern is the effective area of each mode. (The area shown in the blue of the following figure is output area.) In a pattern (0, 4, 5, 6) that specifies the location, the origin of horizontal is the next pixel of SAV 4th and the origin of vertical is the next line of frame information line. By the register PGTHRU, the background can be selected to "Fixed 0h (pattern 0, 6)" and "Setting PGDATA2 (pattern 4, 5)" and "Through" (imaging data). The figure below shows the details.



PG settings (SLVS)

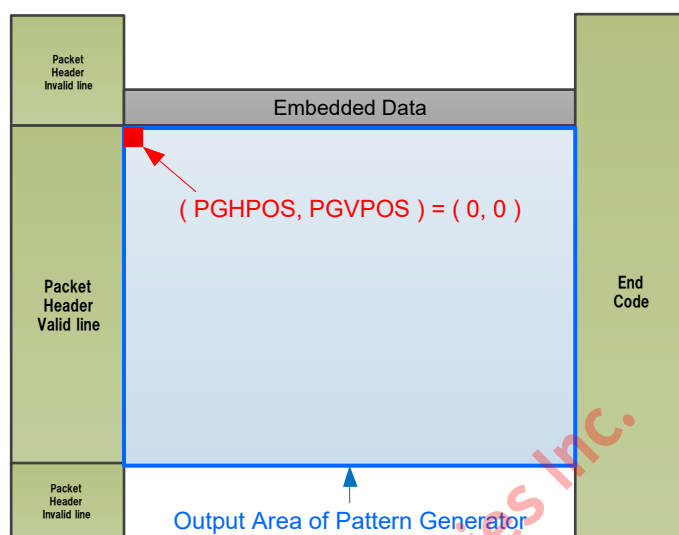
The output range for each output resolution is limited with the value of the following table.

The range of the signal output

Resolution	Output value	
	Min.	Max.
8bit	01h	FEh
10bit	001h	3FEh
12bit	001h	FFEh

8.2.2. When SLVS – EC (Excluding IMX548 series, IMX565 series and IMX568 series) output

The output area of the PG pattern is the effective area of each mode. (The area shown in the blue of the following figure is output area.) In a pattern (0, 4, 5, 6) that specifies the location, the origin of horizontal is the next pixel of Packet Header and the origin of vertical is the next line of Embedded Data line. By the register PGTHRU, the background can be selected to "Fixed 0h (pattern 0, 6)" and "Setting PGDATA2 (pattern 4, 5)" and "Through" (imaging data). The figure below shows the details.



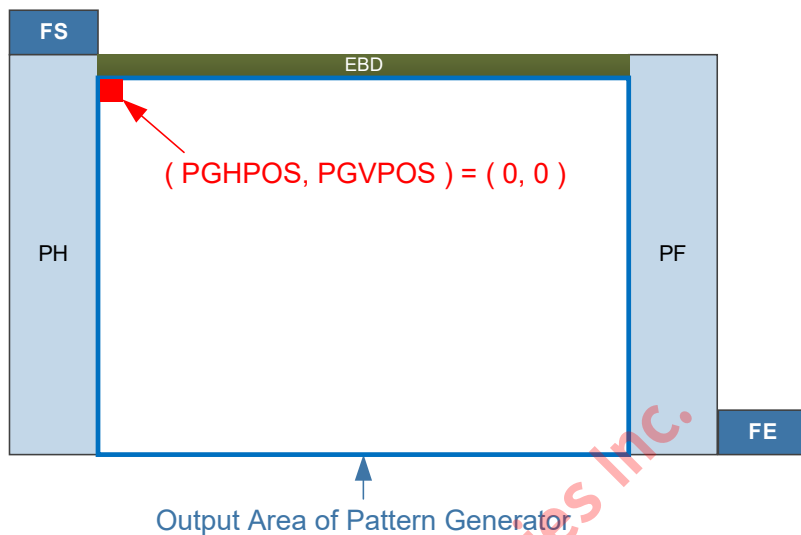
PG settings (SLVS – EC)

The range of the signal output

Resolution	Output value	
	Min.	Max.
8bit	00h	FFh
10bit	000h	3FFh
12bit	000h	FFFh

8.2.3. When CSI-2 (IMX565 series and IMX568 series) output

The output area of the PG pattern is the effective area of each mode. (The area shown in the blue of the following figure is output area.) In a pattern (0, 4, 5, 6) that specifies the location, the origin of horizontal is the next pixel of PH and the origin of vertical is the next line of Embedded Data line (EBD). By the register PGTHRU, the background can be selected to "Fixed 0h (pattern 0, 6)" and "Setting PGDATA2 (pattern 4, 5)" and "Through" (imaging data). The figure below shows the details.



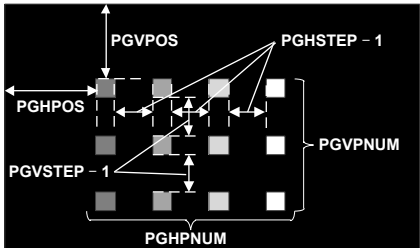
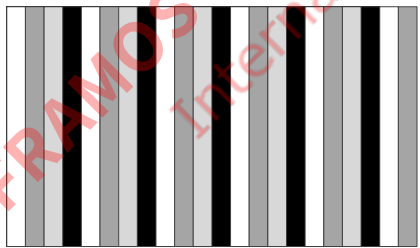
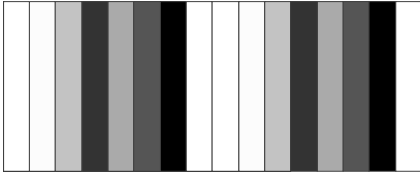
PG settings (CSI-2)

The range of the signal output

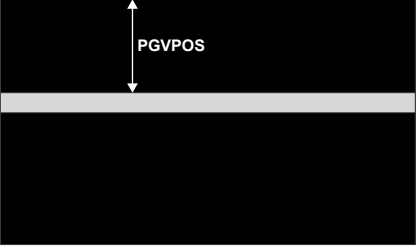
Resolution	Output value	
	Min.	Max.
8bit	00h	FFh
10bit	000h	3FFh
12bit	000h	FFFh

8.3. List of Pattern

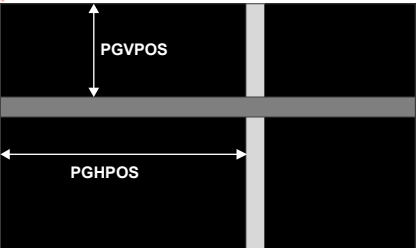
The pattern which it outputs by this function is shown below.

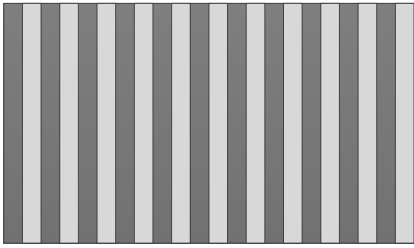
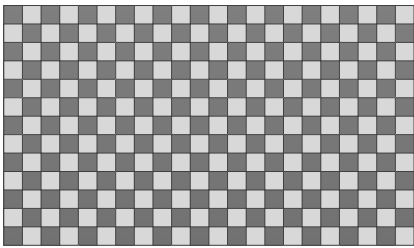
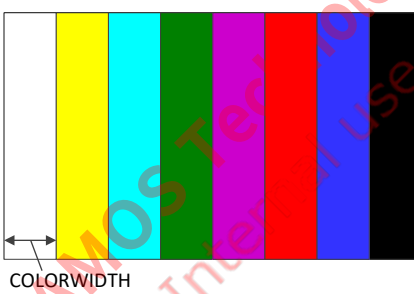
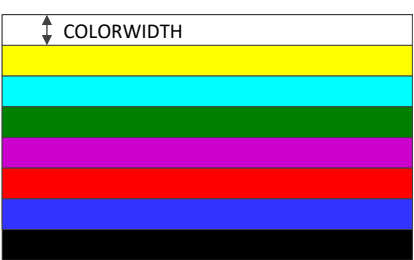
0. Multiple pixels Pattern		
Register	Pattern	Notes
PGMODE = 00h PGDATA1 PGDATA2 PGHPOS PGVPOS PGHPSTEP PGVPSTEP PGHPNUM PGVPNUM		Generate multiple pixels pattern. PGDATA1: Data pattern at the location specified by PGVPOS and PGHPOS The output level of the next point to the right is PGDATA1 + PGDATA2. Since (every each vertical line) The output level is increasing at PGDATA1 + PGDATA2 × n (n = 2, 3, 4...). It clips with the MAX value and returns to the startup level. PGVPOS: Specify the vertical start address PGHPOS: Specify the horizontal start address PGHPSTEP: Interval of horizontal pixels (0h is prohibited) PGVPSTEP: Interval of vertical pixels (0h is prohibited) PGHPNUM: Number of horizontal pixels (0h is prohibited) PGVPNUM: Number of vertical pixels (0h is prohibited)
1. Sequence Pattern 1		
Register	Pattern	Notes
PGMODE = 01h		Outputs the sequential pattern of 1 pixel width 12 bit: FFFh / 555h / AAAh / 000h 10 bit: 3FFh / 155h / 2AAh / 000h 8 bit: FFh / 55h / AAh / 00h
2. Sequence Pattern 2		
Register	Pattern	Notes
PGMODE = 02h		Outputs the following sequential pattern 12bit: FFFh / FFFh / FC0h / FC0h / C3Ch / C3Ch / 333h / 333h / AAAh / AAAh / 555h / 555h / 000h / 000h / FFFh / FFFh 10bit: 3FFh / 3FFh / 3F0h / 3F0h / 30Fh / 30Fh / 0CCh / 0CCh / 2AAh / 2AAh / 155h / 155h / 000h / 000h / 3FFh / 3FFh 8bit: FFh / FFh / FCh / FCh / C3h / C3h / 33h / 33h / AAh / AAh / 55h / 55h / 00h / 00h / FFh / FFh

3. Gradation Pattern		
Register	Pattern	Notes
PGMODE = 03h PGHGSTEP		Outputs the following gradation pattern Minimum to Maximum* It reaches the max and increases from 000h again. * Maximum : 12 bit: 4095d 10 bit: 1023d 8 bit: 255d PGHGSTEP: increment 00b: +1 / 01b: +2 / 10b: +4 / 11b: prohibited

4. Horizontal 1 Row Pattern		
Register	Pattern	Notes
PGMODE = 04h PGDATA1 PGDATA2 PGVPOS		Outputs 1 row pattern with 1 pixel width at registered address. (Valid in the effective pixel region) PGDATA1: Specify the output value PGDATA2: Specify the background value PGVPOS: Specify the vertical address

5. Vertical 1 Column Pattern		
Register	Pattern	Notes
PGMODE = 05h PGDATA1 PGDATA2 PGHPOS		Outputs 1 column pattern with 1 pixel width at registered address. (Valid in the effective pixel region) PGDATA1: Specify the output value PGDATA2: Specify the background value PGHPOS: Specify the horizontal address

6. Horizontal 1 Row and Vertical 1 Column Pattern		
Register	Pattern	Notes
PGMODE = 06h PGDATA1 PGDATA2 PGHPOS PGVPOS		Outputs 1column/row pattern with 1 pixel width at registered address. (Valid in the effective pixel region) PGDATA1: Specify the output value of the column. PGDATA2: Specify the output value of the row. PGHPOS: Specify the horizontal address for column. PGVPOS: Specify the vertical address for row.

7. Stripe Pattern of the arbitrary value		
Register	Pattern	Notes
PGMODE = 07h PGDATA1 PGDATA2		Outputs the vertical stripe pattern of the arbitrary value by 1-pixel width. PGDATA1: Specify the output value PGDATA2: Specify the output value
8. Checks pattern of the arbitrary value		
Register	Pattern	Notes
PGMODE = 08h PGDATA1 PGDATA2		Outputs the checkered pattern of the one pixel spacing with the arbitrary value. PGDATA1: Specify the output value PGDATA2: Specify the output value
9. Color bar which changes horizontally		
Register	Pattern	Notes
PGMODE = 0Ah COLORWIDTH		Outputs the color bar which changes horizontally. COLORWIDTH: Color bar width setting *) 00h and 01h are prohibited COLORWIDTH × 16 pixel unit
10. Color bar which changes vertically		
Register	Pattern	Notes
PGMODE = 0Bh COLORWIDTH		Outputs the color bar which changes vertically. COLORWIDTH: Color bar width setting *) 00h and 01h are prohibited COLORWIDTH × 16 pixel unit

8.4. Test pattern output for SLVS – EC (Excluding IMX548 series, IMX565 series and IMX568 series)

The sensors support test pattern output for SLVS – EC.

Refer to “Annex H. Test Pattern in SLVS – EC Specification Version 2.0” for details.

The test pattern output procedure is shown below.

- (1) CID=0Ch、Address0x0001 = 02h
- (2) CID=0Ch、Address0x0025 = 03h
- (3) CID=0Ch、Address0x0028 = 33h
- (4) CID=0Ch、Address0x0029 = 33h
- (5) CID=0Ch、Address0x002A = 33h
- (6) CID=0Ch、Address0x002B = 13h
- (7) CID=0Bh、Address0x00C1 = 11h
- (8) CID=0Ch、Address0x002B = 93h

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9. How to get sensor information

Following sensors can be confirmed the sensor information of Type name and monochrome (AAM) / color (AAQ) by reading the values of the register as shown below.

Register access is possible after power-on, standby cancel and wait for time^{*1}.

The details are referred to "Sensor Setting Flow" of datasheet.

^{*1}Wait time for each interface.

when SLVS IF is used, wait for 6.8 ms or more.

when SLVS-EC IF is used, wait for longer of "6.8 ms or more" or

"{(SYNCCODE_LEN x 4) + ((DESKEWCODE_INTVL + 4) x DESKEWCODE_LEN)} x (10 / Baud rate) + 2 ^ (INIT_LENGTH + 1) x (10 / Baud rate) + 1.35 ms"

when CSI-2 IF is used, wait for 6.8 ms or more.

9.1. Register map for sensor information

The Register map for sensor information is shown below.

Please refer to the datasheet for registers setup other than those lists.

Chip ID = 0Ah (Write: Chip ID = 0Ah, Read: Chip ID = 8Ah, I²C: 38h)**

Address	bit	Register Name	Description	Reflection timing
16h	5		Type name Information	Read only
	6			
	7			
17h	[6:0]		Monochrome / Color Information	
	7			

9.2. The description of reading data

9.2.1. Type name Information

Data	Type name
1 0 0 0 0 1 0 0 1 0	IMX530
1 0 0 0 0 1 0 0 1 1	IMX531
1 0 0 0 0 1 0 1 0 0	IMX532
1 0 0 0 0 1 0 1 1 1	IMX535
1 0 0 0 0 1 1 0 0 0	IMX536
1 0 0 0 0 1 1 0 0 1	IMX537
1 0 0 0 0 1 1 1 0 0	IMX540
1 0 0 0 0 1 1 1 0 1	IMX541
1 0 0 0 0 1 1 1 1 0	IMX542
1 0 0 0 1 0 0 0 0 1	IMX545
1 0 0 0 1 0 0 0 1 0	IMX546
1 0 0 0 1 0 0 0 1 1	IMX547
1 0 0 0 1 0 0 1 0 0	IMX548
1 0 0 0 1 1 0 1 0 1	IMX565
1 0 0 0 1 1 0 1 1 0	IMX566
1 0 0 0 1 1 0 1 1 1	IMX567
1 0 0 0 1 1 1 0 0 0	IMX568

9.2.2. Monochrome / Color Information

Data	Monochrome / Color
0	Color (AAQ)
1	Monochrome (AAM)

10. Lens design guideline

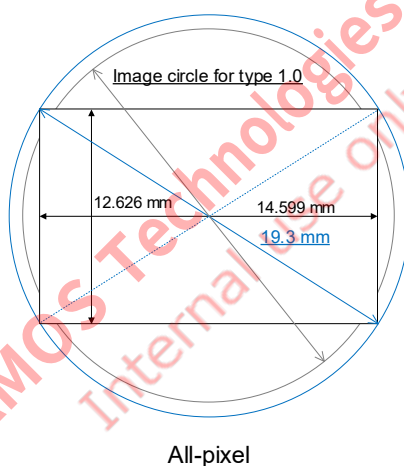
This section describes the information to select the lens.

10.1. Optical system

10.1.1. IMX530 / IMX540 Optical dimension

- ◆ Image size
All-pixel scan mode: Diagonal 19.3 mm (type 1.2)
- ◆ Number of active pixels
All-pixel scan mode: 5328 (H) × 4608 (V) approx. 24.55 M pixels
- ◆ Unit cell size
2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1.0 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)



Relation between Image Circle and the Pixel Area

10.1.2. IMX531 / IMX541 Optical dimension

◆ Image size

All-pixel scan mode: Diagonal 17.5 mm (type 1.1)

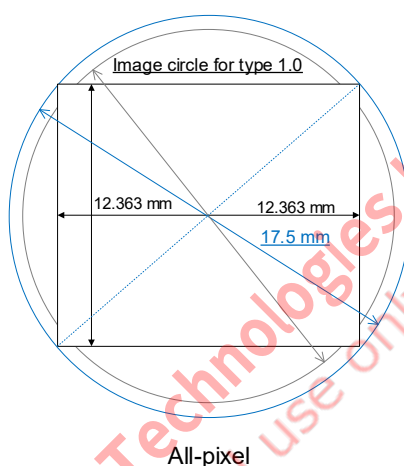
◆ Number of active pixels

All-pixel scan mode: 4512 (H) × 4512 (V) approx. 20.35 M pixels

◆ Unit cell size

2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1.0 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)



Relation between Image Circle and the Pixel Area

10.1.3. IMX532 / IMX542 Optical dimension

◆ Image size

All-pixel scan mode: Diagonal 16.8 mm (type 1.1)

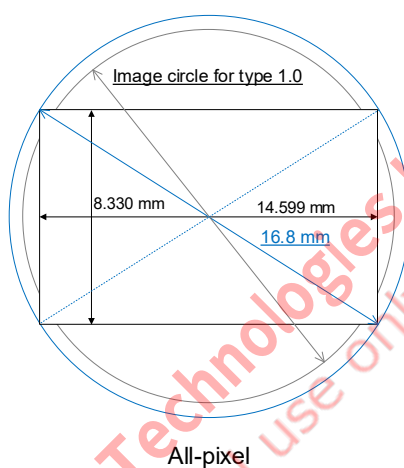
◆ Number of active pixels

All-pixel scan mode: 5328 (H) × 3104 (V) approx. 16.19 M pixels

◆ Unit cell size

2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1.0 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)



Relation between Image Circle and the Pixel Area

10.1.4. IMX535 / IMX545 / IMX565 Optical dimension

◆ Image size

All-pixel scan mode: Diagonal 14.0 mm (type 1/1.1)

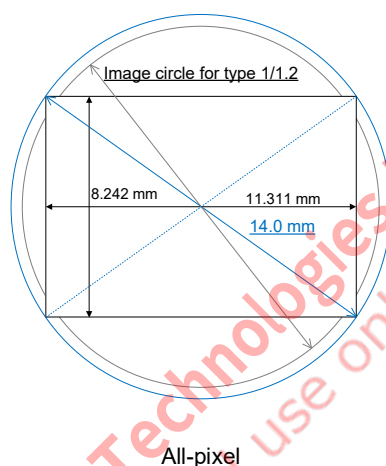
◆ Number of active pixels

All-pixel scan mode: 4128 (H) × 3008 (V) approx. 12.41 M pixels

◆ Unit cell size

2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1/1.2 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)



Relation between Image Circle and the Pixel Area

10.1.5. IMX536 / IMX546 / IMX566 Optical dimension

◆ Image size

All-pixel scan mode: Diagonal 11.1 mm (type 2/3)

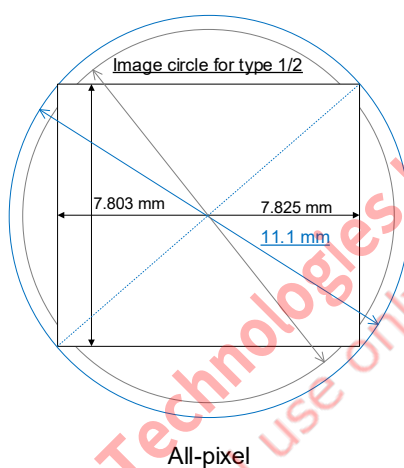
◆ Number of active pixels

All-pixel scan mode: 2856 (H) × 2848 (V) approx. 8.13 M pixels

◆ Unit cell size

2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1/2 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)

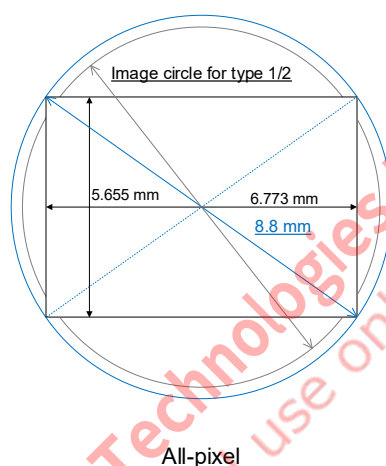


Relation between Image Circle and the Pixel Area

10.1.6. IMX537 / IMX547 / IMX548 / IMX567 / IMX568 Optical dimension

- ◆ Image size
All-pixel scan mode: Diagonal 8.8 mm (type 1/1.8)
- ◆ Number of active pixels
All-pixel scan mode: 2472 (H) × 2064 (V) approx. 5.10 M pixels
- ◆ Unit cell size
2.74 μm (H) × 2.74 μm (V)

Image formation on this image sensor with the lens for type 1/2 is shown below. Considering the color process (de-mosaic etc.) of the image, we recommend that the marginal area for color process is also inside of the image circle. (The necessary margin for color process depends on the algorithm of the de-mosaic. Please investigate the optimum number of the pixels.)



Relation between Image Circle and the Pixel Area

10.2. CRA Characteristics

CRA (Chief Ray Angle) indicates the optimum angle of the chief ray for the image height, independent from the aperture size of the lens.

This sensor does not perform pupil correction. The chief ray angle is assumed to be parallel ray (vertical to sensor image plane).

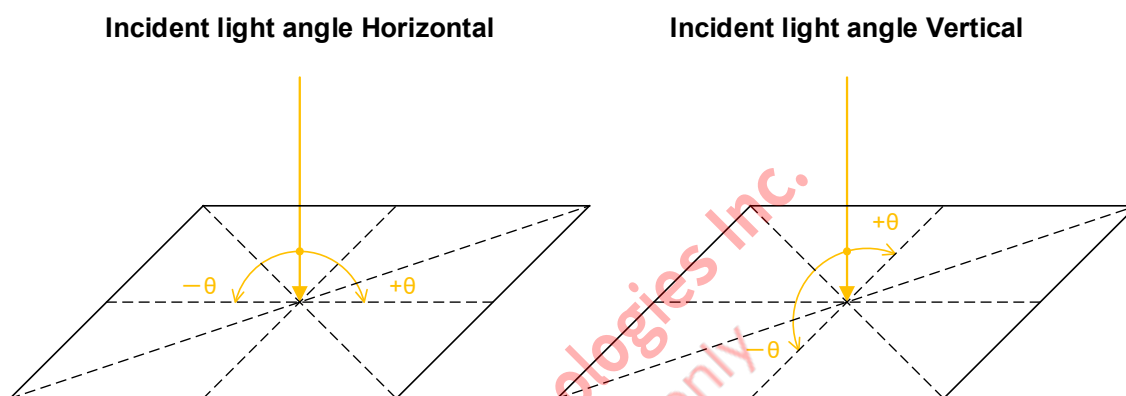
The horizontal and vertical shading characteristics (Reference value) for incident chief rays are shown below.

Register map for sensor information is shown below.

Please refer to the datasheet for registers setup other than those lists.

10.2.1. Measurement method

◆ Incident light angle direction



◆ Measurement method

Enter parallel light and set A and B as follows.

A : The output signal at 0° of incidence angle.

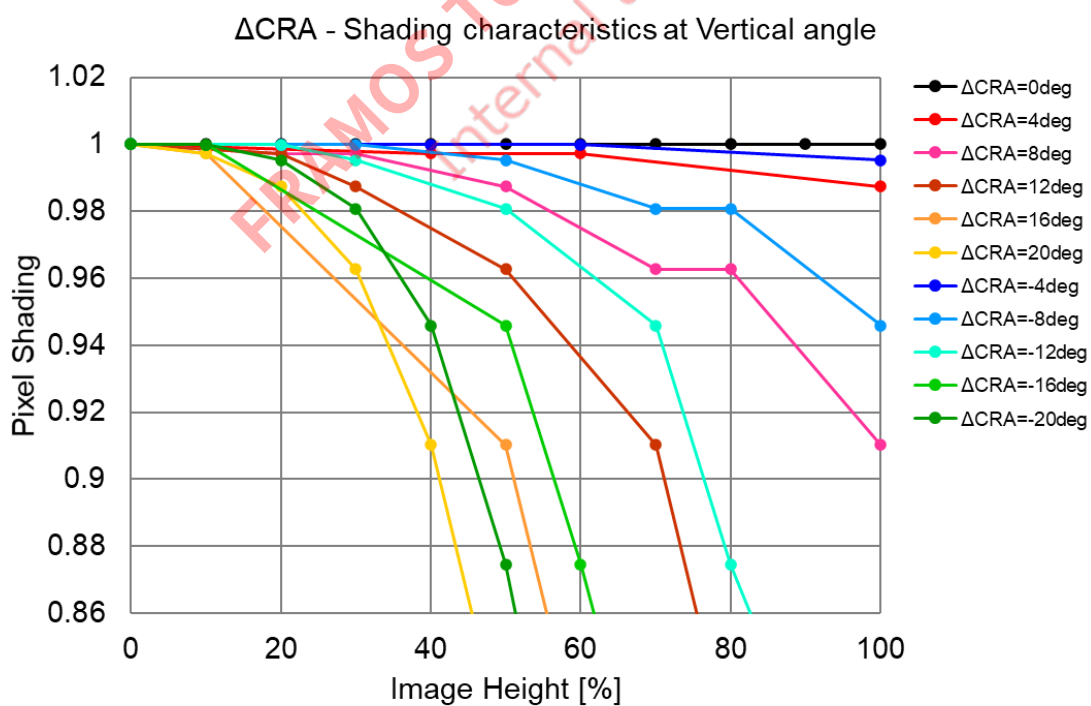
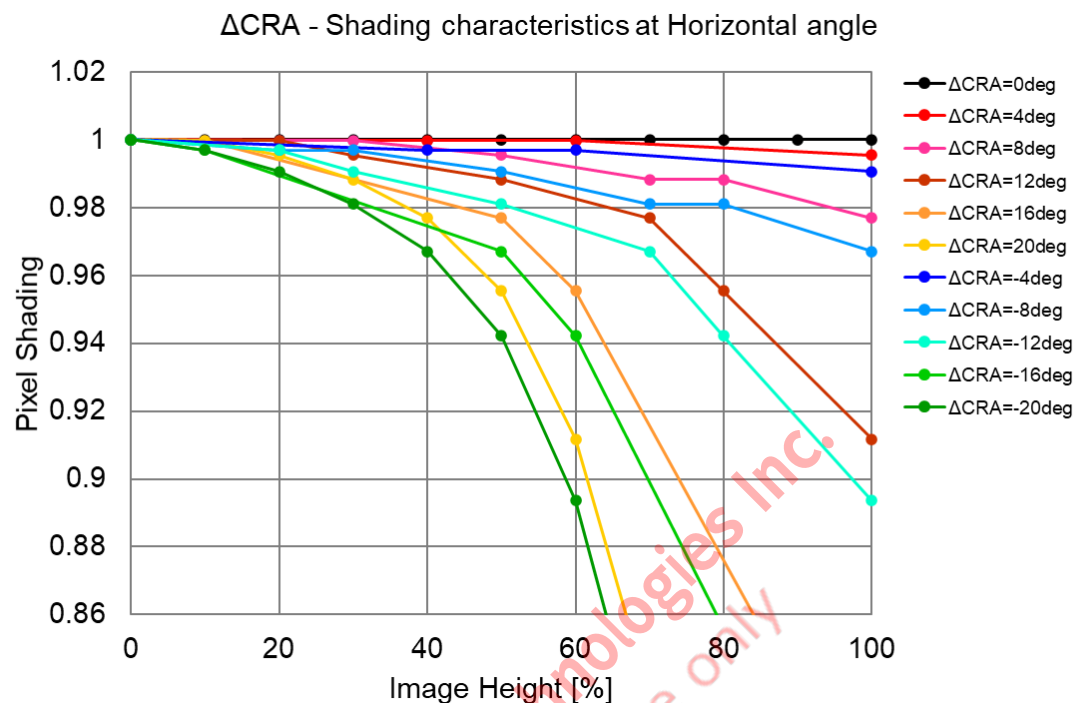
B : The output signal in each degree of incident angle. (with COS correction)

A and B were measured as mentioned above and it calculated as a relative value by the following formula.

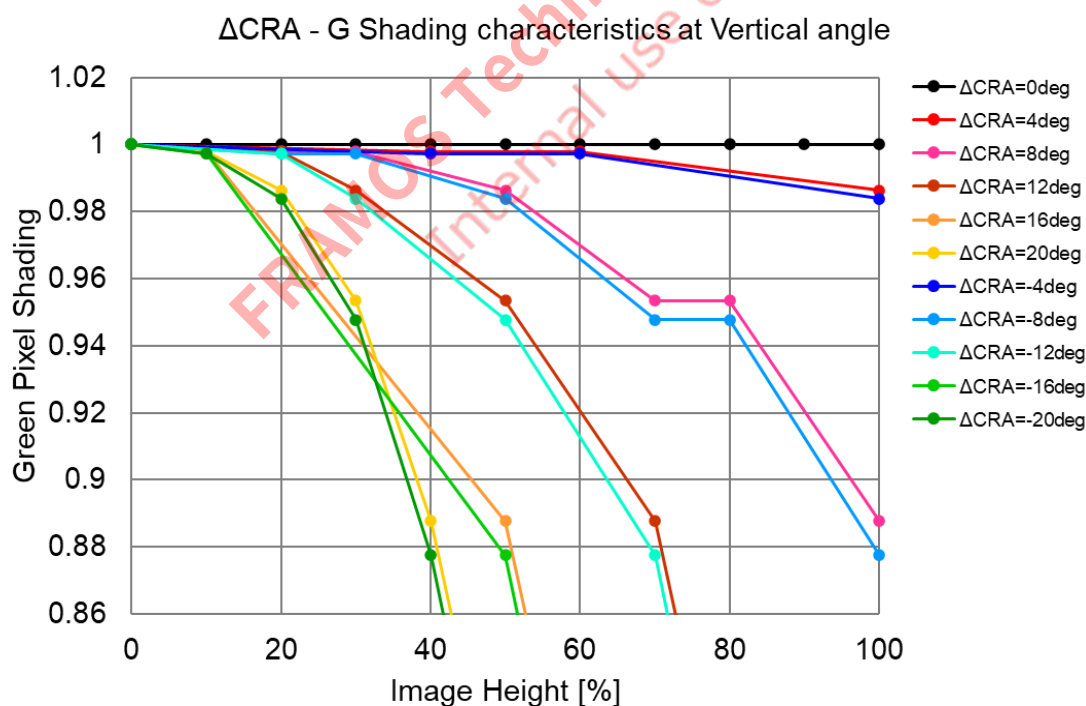
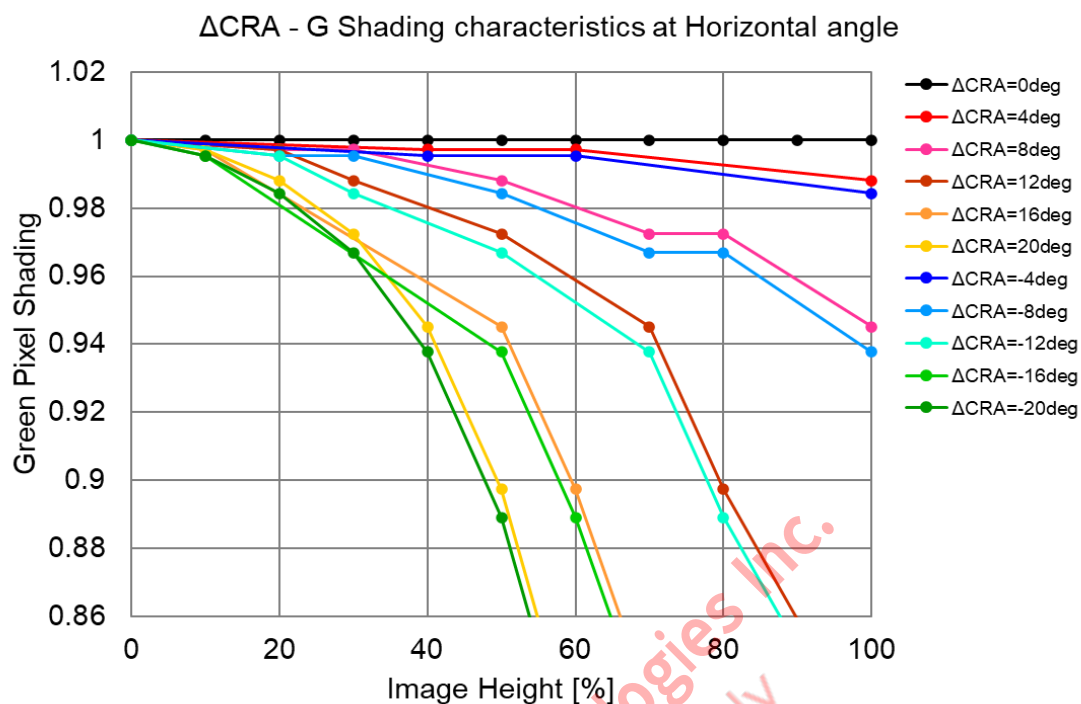
Shading [a.u.] = B / A

10.2.2. Shading characteristics (Reference value)

- ◆ IMX530 series, IMX535 series, IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series monochrome (AAM)

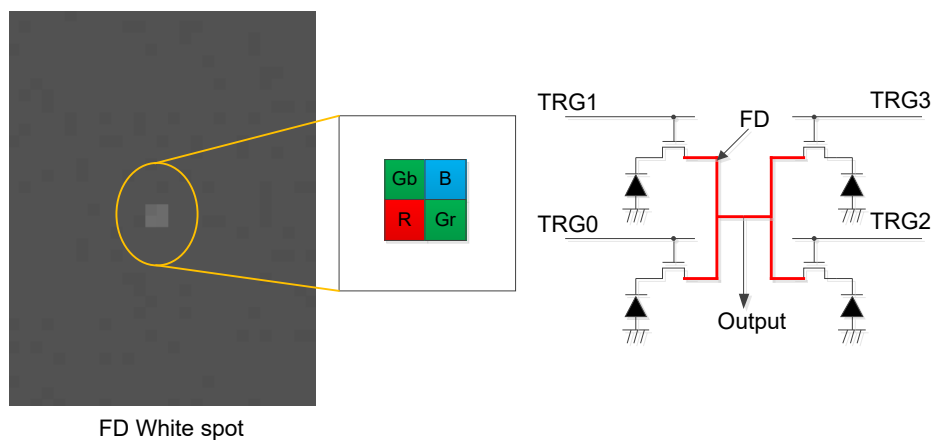


- ◆ IMX530 series, IMX535 series, IMX540 series, IMX545 series, IMX548 series, IMX565 series, IMX568 series color (AAQ)



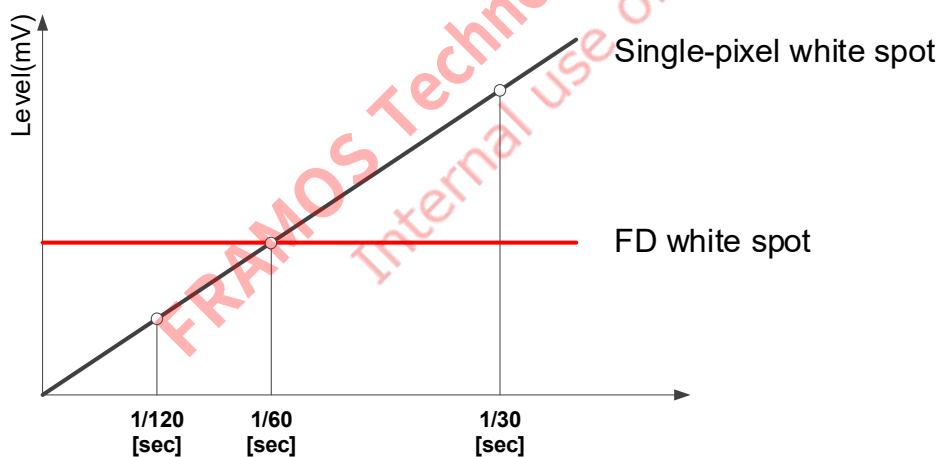
11. FD white spots

This sensor takes the common floating diffusion transistor (FD) architecture to widen the aperture ratio of the pixel. When FD has the leak current or other defect, white spot consist of 4 pixels (2×2 pixels) will appear.



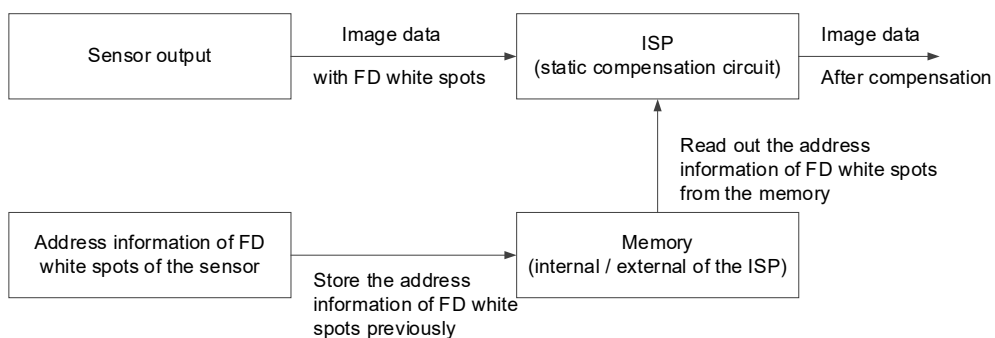
Overview of FD white spot

The signal level of single-pixel white spots (due to the defect of single pixel) increases according to the accumulation time. On the other hand, FD white spot is constant to the accumulation time. There exist the dependency to the temperature, but characteristics are various according to the cause of the spots.



(Both single-pixel white spots and FD white spots are same level at 60 frame/s)

FD white spot is different from single white pixel, cluster spots, then usual dynamic compensation might not be able to conceal it. Static compensation will be necessary. Please consider to use the static compensation of the ISP.



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12. Revision History

Date of change	Revision	Page	Contain of change
2019 / 04 / 02	0.1	–	First edition
2019 / 11 / 20	0.2	All pages	Add the sensor types (IMX531, IMX532, IMX540, IMX541, IMX542, IMX535)
		3	Delete redundant lines of K12 pin
		5	Correction: VRMTRX pins connect to common GND
		8	Delete: Chapter 3.3.1 (3) "Following drawing is ..."
		9	Add the N.C. mark Add the "Example of IMX530" in the figure
		11	Correction: "Standard" of SLVS – EC is changed to "Specification".
		14	Update: "version" of SLVS – EC Specification is changed to "Baud Grade" Update: "MOU" is changed to "License Agreement"
		15	Delete the version for SLVS – EC
		17, 18	Correction: register Chip ID is changed to "for read / for write"
		21	Delete: the pin number
		24	Correction: Figure where input line and output line of SDA were separated
		24	Correction: initial value of PGCLKEN
		26	Correction: initial value of PGHPSTEP
		26	Correction: initial value of PGVPSTEP
		33	Support the sensor identification function Add the sensor types (IMX531, IMX532, IMX540, IMX541, IMX542, IMX535) Add the IMX535 is not yet support in the note
		34	Correction: "recording pixel" is changed to "Active pixel"
		34 to 39	Add the sensor types (IMX531, IMX532, IMX535, IMX540, IMX541, IMX542, IMX535) in Optical dimension and CRA characteristics
		38	Correction: "The CRA characteristics ... is" is changed to "The CRA characteristics ... are"
2020 / 1 / 6	0.3	All pages	Add: the sensor type (IMX545)
		1	Add: IMX535-AAQJ-C, IMX545-AAMJ-C
		5	Delete: number of the section 3.3
		14	Add: The mark of object of document for which signature of the License Agreement is necessary
		20	Correction: font of I2C (I2C→I ² C)
		33	Add: the sensor type (IMX545) Add the IMX535 series and IMX545 are not yet support in the note
		37, 41	Add the IMX545 in Optical dimension and CRA characteristics

Date of change	Revision	Page	Contain of change
2020 / 5 / 25	1.0	All pages	Add sensor types (IMX535 series, IMX545 series and IMX565 series)
		3, 4	Change the description for LGA package
		5	Add the reference design for IMX565 series
		11, 17	Add the description of CSI-2
		34	Add the Annex.H of SLSI-2
		35	Add the IMX535/545/565 series to sensor information
		39	Add the IMX565 to optical dimension
		40	Add the IMX536/546/566 to optical dimension
		41	Add the IMX537/547/567 to optical dimension
		45	Add IMX565
		46	Add CRA for IMX536/546/566
		47	Add CRA for IMX537/547/567
2020 / 9 / 10	1.1	1	Add IMX548 series
		3,5	Add IMX548
		4, 6, 9	Add IMX548 Applied circuit example
		12	Correction: Power supply pins (3)
2020 / 10 / 28	1.2	1	Add IMX548 series and IMX568 series
		3, 4, 5, 10, 13, 15, 16, 20, 21, 34	Add IMX568 series
		8	Add the reference design for digital power supply pins for IMX568 series
		15, 16, 20, 34	Add IMX548 series
		38	Add IMX548 series / IMX565 series / IMX568 series
		39	Add the type name information for IMX548 and IMX568 Change the note *1 of 9.2.1 and 9.2.2
		45	Add the optical dimension of IMX548 and IMX568
		46	Change the description of CRA characteristics
2020 / 12 / 21	2.0	1	Dalet the (Tentative) word of IMX535 series, IMX545 series, and IMX565 series
		3, 4	Correct the Reference design for analog power supply pins
		34, 35	Correct the 8.2.2 and add the 8.2.3
		39	Correct the address value of (5), (6), and (8)

Date of change	Revision	Page	Contain of change
2021 / 3 / 29	2.1	3 to 10, 15	Delete "LGA package" from the Figure caption
		4 to 8	Correct the Reference design
		11	Add "4.1.3. Notes for the mount"
		22	Change the description of "5.2. CSI-2 Serial Output"
		41	Delete the note "*1 IMX548 series and IMX568 series are not yet suppor" in 9.2.1 and 9.2.2
2021 / 5 / 19	3.0	1	Delete "Tentative" in IMX548 series and IMX568 series
		30	Correct the Default value after reset for Add=50h PGCLKEN
		46	Correct the description of the Image formation
		48 to 50	Add CRA characteristics
2023 / 1 / 5	4.0	6, 8	Correction: VDDLSC from "A1" to "A3"
2023 / 11 / 10	5.0	3	Correct the connect for VDD33SUB and other in figure of Reference design for analog power supply pins.
		4	Correct the connect for VDD33SUB in figure of Reference design for analog power supply pins.
		6	Correct the connect for VDDLSC(N3) and other in figure of Reference design for digital power supply pins.
		8	Correct the connect for VDDLSC(N3) and other in figure of Reference design for digital power supply pins.
		41	Update the wait time for register access.